CardioCog: An Explainable AI Framework for Cognitive Health Assessment and Early Prediction of Cognitive Decline Using Cardiovascular, Genetic, and Clinical Factors

A Thesis

Submitted in partial fulfilment for the Degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

by

Akula Akhil Vighnesh - CB.SC.U4AIE23303

Vada Gouri Hansika Reddy - CB.SC.U4AIE23304

Pilli Venkata Bhargav Reddy - CB.SC.U4AIE23329

Marella Venkata Mahidhar Reddy - CB.SC.U4AIE23354



DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS (INDIA)

October 2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS



BONAFIDE CERTIFICATE

This is to certify that the thesis entitled "**CardioCog: An Explainable AI Framework for Cognitive Health Assessment and Early Prediction of Cognitive Decline Using Cardiovascular, Genetic, and Clinical Factors**" submitted by **Akula Akhil Vighnesh, CB.SC.U4AIE23303**, **Vada Gouri Hansika Reddy, CB.SC.U4AIE23304**, **Pilli Venkata Bhargav Reddy, CB.SC.U4AIE23329** and **Marella Venkata Mahidhar Reddy, CB.SC.U4AIE23354** for the award of the **Degree of Bachelor of Technology** under the **Faculty of Engineering** is a bonafide record of the work carried out by them under our guidance and supervision at "Department of Artificial Intelligence", Amrita Vishwa Vidyapeetham, Coimbatore Campus.

Submitted for the examination held on 28 October 2026.

INTERNAL EXAMINER

Dr. Manoj Bhat K
Thesis Advisor,
Assistant Professor,
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

EXTERNAL EXAMINER

Amrutha Veluppal
Thesis Co-Advisor,
Assistant Professor,
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS

DECLARATION

We, **Akula Akhil Vighnesh, CB.SC.U4AIE23303**, **Vada Gouri Hansika Reddy, CB.SC.U4AIE23304**, **Pilli Venkata Bhargav Reddy, CB.SC.U4AIE23329** and **Marella Venkata Mahidhar Reddy, CB.SC.U4AIE23354** hereby declare that this thesis entitled "**CardioCog: An Explainable AI Framework for Cognitive Health Assessment and Early Prediction of Cognitive Decline Using Cardiovascular, Genetic, and Clinical Factors**" is the record of the original work done by us under the guidance of **Dr. Manoj Bhat K**, Assistant Professor, and the co-guidance of **Amrutha Veluppal**, Assistant Professor, Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus. To the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/associateship/fellowship or a similar award to any candidate in any University.

Place: Coimbatore, India **Signature of the Students**
Date: 28 October 2026

COUNTERSIGNED

Dr. K.P Soman
Professor and Dean,
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

Contents

Acknowledgements **v**

List of figures **vi**

List of tables **ix**

Abbreviations **x**

List of symbols **xi**

Abstract **xii**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 How the Research Question Developed 1

1.3 Changes from the Initial Proposal 2

1.3.1 The First-Review Prototype 3

1.4 Problem Statement 4

1.5 Research Questions and Objectives 4

1.6 Contributions 5

1.7 Organisation of the Thesis 6

2 Literature Review **7**

2.1 Dementia, Mild Cognitive Impairment and Their Definition 7

2.2 Modifiable and Cardiovascular Risk Factors 7

2.3 Genetic Risk 8

2.4 Amyloid and Tau PET 8

2.5 Machine Learning for Tabular Clinical Data 8

2.6 Explainability 9

2.7 Retrieval-Augmented Generation 9

2.8 Related Prediction Studies . . . 9
2.9 Data Source . . . 10
2.10 Research Gap . . . 10

3 Data **11**
3.1 The National Alzheimer's Coordinating Center . . . 11
3.2 Files received . . . 11
3.3 The clinical table . . . 12
3.3.1 Missing-value codes . . . 13
3.3.2 Changes in the forms . . . 13
3.4 Feature categories . . . 14
3.5 PET data . . . 14

4 Methodology **15**
4.1 Prediction task . . . 15
4.1.1 Choice of horizon and outcome . . . 15
4.1.2 Censoring . . . 16
4.1.3 Secondary outcomes: cognitive decline . . . 17
4.2 Cohort . . . 17
4.3 Feature construction . . . 18
4.3.1 Harmonised cognitive scores . . . 18
4.3.2 Cardiovascular history . . . 19
4.3.3 Genetics . . . 19
4.3.4 Preventing the use of future information . . . 19
4.4 Data splits . . . 20
4.5 Models . . . 20
4.5.1 Mathematical form of the models . . . 21
4.6 Modality-Encoder Fusion Network . . . 22
4.7 Contribution of each feature category . . . 23
4.8 Longitudinal models . . . 24
4.9 PET late fusion . . . 24
4.10 Modality dropout . . . 24
4.11 Explainability . . . 25
4.12 Evaluation . . . 25
4.12.1 Survival analysis . . . 26

5 PET Image Pipeline **27**

5.1 From DICOM series to static volumes . . . 27
5.2 Spatial registration . . . 27
5.3 Amyloid SUVR from the images . . . 29
5.4 Amyloid status classification . . . 30
5.5 Grad-CAM . . . 31
5.6 Why images are not used to predict dementia . . . 32

6 Results **34**
6.1 Comparison of models . . . 34
6.2 Performance by diagnosis at the index visit . . . 35
6.3 What drives the prediction . . . 36
6.3.1 Feature categories . . . 36
6.3.2 Cardiovascular information (research question 2) . . . 36
6.3.3 Shapley explanations . . . 37
6.4 Cognitive Decline as the Outcome . . . 38
6.4.1 Predicted Change in CDR-SB . . . 39
6.5 Modality-Encoder Fusion Network . . . 40
6.6 Longitudinal information . . . 40
6.7 PET measures (research question 3) . . . 41
6.8 Validation . . . 43
6.8.1 Subgroups . . . 43
6.8.2 Calibration and clinical usefulness . . . 43
6.9 Missing information . . . 44
6.10 Sensitivity analyses . . . 45

7 System Design **47**
7.1 Architecture . . . 47
7.2 Input form . . . 47
7.3 Output . . . 48
7.4 Plain-Language Report . . . 49
7.5 Plausibility checks . . . 50
7.6 Behaviour checks . . . 50
7.7 Evidence assistant . . . 50

8 Verification and Reproducibility **53**
8.1 Checks . . . 54
8.2 Errors found and corrected . . . 54

8.3 Reproducibility 55

9 Discussion and Limitations **56**

9.1 Interpretation of the Main Result 56

9.2 The Cardiovascular Question 56

9.3 Comparison with Related Studies 57

9.4 Model Complexity 57

9.5 PET 57

9.6 Limitations 58

9.7 Ethical Considerations 58

10 Conclusion and Scope for Further Research **60**

10.1 Conclusion 60

10.2 Scope for Further Research 61

10.2.1 External and Temporal Validation 61

10.2.2 Cognitively Normal Participants and Longer Horizons 61

10.2.3 Richer Cardiovascular Exposure 61

10.2.4 PET with Longer Follow-up 61

10.2.5 Evidence Assistant 61

10.3 Final Remarks 62

A Feature List **63**

B Implementation and Reproducibility Notes **64**

B.1 Software 64

B.2 Pipeline 64

B.3 Seeds 65

B.4 Data Access 65

References **66**

Acknowledgements

We express our sincere gratitude to our thesis advisor, Dr. Manoj Bhat K, and our co-advisor, Amrutha Veluppal, for their guidance and constructive feedback throughout this work. We thank the Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus, for the academic environment and facilities that made this project possible, and the faculty members and review panel for their suggestions on the research design and evaluation. We are grateful to our families and friends for their support.

This work uses data from the National Alzheimer's Coordinating Center (NACC). We thank the NACC, the Alzheimer's Disease Research Centers that contribute to it, and above all the research participants and their study partners.

AKULA AKHIL VIGHNESH
VADA GOURI HANSIKA REDDY
PILLI VENKATA BHARGAV REDDY
MARELLA VENKATA MAHIDHAR REDDY

List of Figures

1.1 Overview of the work presented in this thesis. 6

3.1 Percentage of in-person visits with a valid score for each neuropsychological test, by UDS version. Tests of versions 1–2 and of version 3 are almost disjoint. 13

4.1 Kaplan–Meier estimates of cumulative progression from the first visit: cognitively normal participants to MCI or dementia, and participants with MCI to dementia. 16

4.2 Definition of the outcome. Only the index visit supplies features; later visits supply the label. 16

4.3 Demographically adjusted z-score for the MoCA, using the fitted normative model. 18

4.4 Left: mean domain z-score of cognitively normal participants not used to fit the norms, by UDS version. Right: mean domain z-score by diagnosis at the first visit. 19

4.5 Participant-level partitions of the main cohort. 20

5.1 Quality control of the conversion and registration: number of frames and minutes of data averaged per scan, and the volume scaling applied by the affine registration (1 = no change). 28

5.2 Average of all registered scans for each tracer class (amyloid, 318 scans; tau, 232; FDG, 49), in axial, coronal and sagittal sections. In the top row the orange outline is the Centiloid cortical target region and the green outline the cerebellar reference region. Only group averages are shown; no individual scan is reproduced. 29

5.3 Image-derived amyloid SUVR against the value computed by the NACC PET core for the same scans. 30

5.4 Cross-validated AUROC with 95% bootstrap intervals for the three amyloid-status classifiers. 31

5.5 Group-average SUVR images of amyloid-negative and amyloid-positive scans (top two rows) and the mean Grad-CAM map of the 3D CNN (bottom row), in four sections. The blue outline marks the standard cortical target region. 32

6.1 AUROC with 95% bootstrap intervals of the five models on the test set and the held-out centres (labelled "external set"). 35

6.2 Left: AUROC of each category alone and added cumulatively. Right: gain in AUROC from adding each category to all the others. 36

6.3 The fifteen features with the largest mean absolute SHAP value on the test set, coloured by category. The grey text gives the range of each feature's rank over eight models retrained on bootstrap samples. 37

6.4 Share of SHAP importance by category within cognitively normal participants and participants with MCI at the index visit. 38

6.5 AUROC with 95% intervals of XGBoost for dementia within three years and for a rise of at least one point in CDR-SB within three years, by diagnosis at the index visit. 39

6.6 Mean absolute error of the predicted three-year change in CDR-SB, against the reference of predicting no change, by diagnosis at the index visit. . . . 39

6.7 Mean attention share of each modality in the fusion network, by evaluation set and diagnosis at the index visit. The dashed line marks uniform attention (20%). 40

6.8 AUROC with 95% intervals of the three longitudinal models on landmark visits. 41

6.9 Percentage of participants with MCI at the index visit who converted to dementia within two and three years, by amyloid status, under the strict and nearest scan-matching rules (test set and held-out centres). 42

6.10 Change in AUROC when PET measures are added to the clinical model, with 95% intervals, test set and held-out centres pooled. The number of events is given for each comparison. 42

6.11 XGBoost AUROC with 95% intervals by subgroup on the test set and the held-out centres ("external set"). 43

6.12 Calibration of XGBoost: observed share converting against mean predicted risk in groups of participants, on the test set and the held-out centres. . . . 44

6.13 Decision curves: net benefit of acting on the model's prediction, of acting on everyone and of acting on no one, across risk thresholds. 44

6.14 AUROC and Brier score on the test set when categories of information are missing, for the standard model, the modality-dropout model and a specialist model retrained for each pattern. 45

6.15 Hazard ratios per standard deviation with 95% intervals from the Cox model, adjusted for all other features. Cardiovascular and medical features are shown in red. 45

7.1 The estimate panel of the prototype for the example patient: estimated risk, observed rate of similar participants, categories used, SHAP contributions by category, test scores relative to healthy people of the same age, sex and education, and the most influential single items. 48

7.2 Plain-language report generated by the prototype for the example patient. 49

7.3 The evidence assistant answering “Is midlife hypertension associated with later dementia?” with quoted passages and numbered sources. 51

7.4 Share of the 32 in-scope questions for which the top-ranked passage came from an expected source, for keyword retrieval, embedding retrieval and the hybrid. 52

List of Tables

1.1 Components proposed at the zeroth review and what was implemented. . . 3

3.1 Files in the NACC release and their role in this study. 12
3.2 Feature categories, source columns and number of model features. 14

4.1 Cohort flow. 17
4.2 Partition sizes for the main cohort. 20
4.3 Models and settings. 21

5.1 Agreement between image-derived and NACC amyloid SUVR (318 scans). 30
5.2 Amyloid status classification (292 scans, 84 positive, cross-validated). . . . 31

6.1 Discrimination and calibration of the five models. 34
6.2 XGBoost by diagnosis at the index visit (participants / converters). 35
6.3 Contribution of each feature category (XGBoost, test set unless stated). . . 36
6.4 Share of mean absolute SHAP value by category (test set, 2,286 participants). 37
6.5 XGBoost for CDR-SB worsening of at least one point within three years (participants / events). 38
6.6 Effect of earlier visits (paired gain over the landmark-only model). 41
6.7 XGBoost AUROC by subgroup. 43
6.8 Brier score when categories are missing (test set; lower is better). 44

7.1 Components of the prototype. 47
7.2 Evaluation of the evidence assistant (52 questions). 51

8.1 Independent verification checks and their results. 54

A.1 The 48 model features and their source columns in the NACC clinical table. 63

Abbreviations

APOE	–	Apolipoprotein E
AUPRC	–	Area Under the Precision–Recall Curve
AUROC	–	Area Under the Receiver Operating Characteristic Curve
BM25	–	Best Matching 25 (keyword ranking function)
BMI	–	Body Mass Index
CDR	–	Clinical Dementia Rating
CDR-SB	–	Clinical Dementia Rating, Sum of Boxes
CI	–	Confidence Interval
CNN	–	Convolutional Neural Network
DICOM	–	Digital Imaging and Communications in Medicine
FAQ	–	Functional Activities Questionnaire
FDG	–	Fluorodeoxyglucose
FT-Transformer	–	Feature Tokenizer Transformer
GDS	–	Geriatric Depression Scale
GRU	–	Gated Recurrent Unit
MCI	–	Mild Cognitive Impairment
MLP	–	Multilayer Perceptron
MMSE	–	Mini-Mental State Examination
MoCA	–	Montreal Cognitive Assessment
NACC	–	National Alzheimer's Coordinating Center
NPI-Q	–	Neuropsychiatric Inventory Questionnaire
PET	–	Positron Emission Tomography
RAG	–	Retrieval-Augmented Generation
SHAP	–	SHapley Additive exPlanations
SUVR	–	Standardised Uptake Value Ratio
UDS	–	Uniform Data Set
XGBoost	–	Extreme Gradient Boosting

List of symbols

$\mathbf{x}_{t_0}$	–	Feature vector at the index visit
t_0	–	Date of the index visit
y, y_{dec}	–	Dementia and CDR-SB decline outcomes within three years
Δ, $\hat{\Delta}$	–	Observed and predicted three-year change in CDR-SB
$\hat{p}$	–	Predicted probability of the outcome
$\sigma(\cdot)$	–	Logistic function, $1/(1+e^{-u})$
s, $\hat{s}$	–	Raw test score and expected score of a cognitively normal person
σ_{res}	–	Residual standard deviation of the normative model
z	–	Demographically adjusted test score
β_k	–	Regression coefficients
C	–	Inverse strength of the L_2 penalty in logistic regression
η, λ	–	Learning rate and leaf-weight penalty of XGBoost
$\mathbf{t}_g$, $\mathbf{m}_g$	–	Token and missing token of modality g in the fusion network
α_g	–	Attention weight of modality g
$\rho(\cdot)$	–	Huber loss
ξ	–	Modality-dropout indicator, Bernoulli(0.35)
ϕ_j	–	SHAP contribution of feature j
ΔAUROC	–	Paired difference in AUROC between two models
B	–	Brier score
NB(t)	–	Net benefit at risk threshold t
$\hat{S}(t)$, $h(t)$	–	Kaplan–Meier survival estimate and hazard function
$V(\mathbf{r})$	–	PET activity at voxel $\mathbf{r}$
SUVR	–	Standardised uptake value ratio
MI	–	Mutual information between two images

Abstract

Many older adults who do not have dementia today will develop it within a few years, yet the information that clinics routinely collect about them is rarely combined into a single, explainable estimate of that risk. This thesis presents CardioCog, a framework that predicts cognitive decline over the next three years from demographic, cardiovascular, genetic and clinical information, adds amyloid and tau PET when it is available, explains each prediction, and supports it with published evidence.

The work uses the National Alzheimer's Coordinating Center database (57,038 participants, 217,598 visits). Neuropsychological tests from different form versions were placed on one scale with demographically adjusted z-scores, and 48 features in five categories were built for every participant. Cognitive decline was defined in two ways: progression to a diagnosis of dementia (19,470 participants, 2,818 events) and a worsening of at least one point in the CDR Sum of Boxes (20,355 participants, 5,035 events), together with the predicted three-year change in that score. Participants were split by person, and nine whole centres were held out for evaluation.

Five classifiers, from logistic regression to an FT-Transformer, and a modality-encoder fusion network with attention all performed alike: AUROC 0.93 to 0.94 for dementia (0.94 to 0.95 at held-out centres) and about 0.86 for CDR-SB decline (0.89). Within participants with mild cognitive impairment the AUROC was 0.84 for dementia and 0.75 for decline. Most of the predictive signal came from current cognitive status. Cardiovascular and medical history added about 0.001 AUROC for dementia and nothing measurable for decline once cognition was known; earlier visits added at most 0.005, and PET measures produced no demonstrable gain. Training with modality dropout kept predictions calibrated when whole categories were missing. A separate PET image pipeline reproduced the PET core's amyloid measurements (correlation 0.98 over 318 scans). The models were built into a research prototype that checks the plausibility of its inputs and writes a plain-language report with quoted evidence, and the pipeline passed seventeen independent verification checks, which also found and corrected two sources of information leakage.

Keywords: dementia prediction, mild cognitive impairment, NACC, cardiovascular risk, explainable machine learning, SHAP, amyloid PET, retrieval-augmented generation.

Chapter 1

Introduction

1.1 Background and Motivation

The changes in the brain that lead to dementia build up over years, and many people pass through a middle stage, mild cognitive impairment (MCI), in which memory or thinking is measurably weaker but daily life is still managed independently. Some people stay in that stage for a long time, some return to normal, and some go on to dementia within a few years. Even among people who seem cognitively normal, a small number decline quickly. If clinicians could tell, early and reliably, who is heading towards decline, they could follow those people more closely, plan care with the family, and select the right participants for clinical trials.

Much of the information needed for such an estimate is already collected in memory clinics. Age is the strongest single risk factor. Cardiovascular and metabolic conditions such as hypertension and diabetes are associated with later dementia in large epidemiological studies [1–4]. The APOE $\varepsilon 4$ allele is the main genetic risk factor for late-onset Alzheimer's disease [5]. Neuropsychological tests and clinical ratings describe how a person is thinking and functioning today, and amyloid and tau PET imaging show the underlying pathology directly [6, 7]. What remains unclear is how much each of these sources really adds once the others are known, and whether complex machine-learning models are needed to combine them.

This project, CardioCog, set out to build an explainable framework that assesses cognitive health and predicts cognitive decline from cardiovascular, genetic and clinical information, with PET where it is available, and to support each prediction with published evidence.

1.2 How the Research Question Developed

The project began with a broad aim: to combine cardiovascular risk factors, cognitive trajectories, APOE and PET biomarkers to predict cognitive decline, and to explain the

predictions with supporting literature. Before any model was trained, the data were audited in detail. Three findings from that audit determined the design of the study. First, the clinical diagnosis and the Clinical Dementia Rating (CDR) are recorded at every visit across twenty years of data, whereas the neuropsychological battery changed in 2015. Second, roughly one in three participants with MCI progresses to dementia within three years, so a three-year window contains enough events to learn from. Third, most PET scans are recent, and only a few hundred participants have raw images, of whom very few have declined since.

"Cognitive decline" was therefore made concrete in two complementary ways, both measured over three years from a clinic visit. The primary outcome is progression to a clinical diagnosis of dementia. The secondary outcome is a worsening of at least one point in the CDR Sum of Boxes (CDR-SB), which also captures people who decline without reaching dementia; its continuous counterpart, the change in CDR-SB, is used for a progression estimate. The original cardiovascular hypothesis was kept, but as a question to be tested rather than an assumption.

1.3 Changes from the Initial Proposal

The framework presented at the zeroth review was designed before the data had been examined, and the first review reported an early imaging prototype. Several components were changed once the data showed what they could support. Table 1.1 records each change and the reason for it.

Table 1.1: Components proposed at the zeroth review and what was implemented.

Proposed	Implemented	Reason
Cognitive-decline risk and progression	Dementia within three years (primary); CDR-SB worsening of at least one point (secondary); three-year CDR-SB change (regression)	Diagnosis and CDR are recorded consistently in every form version; three years balances events against loss to follow-up
NACC (ADSP-PHC release) and NHANES	NACC Freeze 74: clinical data, PET summary tables and raw PET images	This release contains every required variable with longitudinal visits and linked PET data; NHANES does not provide longitudinal dementia diagnoses
Separate encoders per modality, two-tier fusion with attention	Modality-encoder network with attention fusion and missing-modality tokens (Tier 1); PET added by late fusion (Tier 2)	Implemented as proposed; PET is fused separately because too few scanned participants have three-year outcomes to train it inside the network
Risk head and progression head	Classification of both outcomes and regression of CDR-SB change, in XGBoost and in the fusion network	Implemented
XGBoost or CatBoost	Logistic regression, random forest, XGBoost, MLP, FT-Transformer	A wider comparison answers whether model complexity helps; all models performed alike, so CatBoost was not added
LSTM or Transformer over visits, with temporal attention	Trajectory features and a GRU on landmark visits	A GRU suits short, irregular sequences of at most six visits; history added at most 0.005 AUROC, leaving little for visit-level attention to explain
3D ResNet for PET	Validated PET image pipeline and a small 3D CNN for amyloid status	Only 5 of 446 participants with images progressed to dementia, so an image model of decline could not be trained reliably (Section 1.3.1)
SHAP, Grad-CAM, attention weights, DiCE	SHAP, Grad-CAM, modality attention weights	Counterfactual explanations suggest what a patient should change, which observational data cannot support
RAG grounding and plain-language report	Evidence assistant with hybrid retrieval and guardrails; plain-language report with quoted evidence	Implemented
Target AUROC 0.70 to 0.78	0.94 for dementia, 0.86 for CDR-SB worsening; 0.84 and 0.75 within MCI	The overall figures are higher than expected mainly because current cognitive status is a strong predictor, so results are also reported within diagnostic groups

1.3.1 The First-Review Prototype

The work shown at the first review followed the original imaging plan closely. A 3D CNN on raw amyloid PET volumes was fused with 34 clinical features through spatial cross-attention, to predict a rise of at least 0.5 in CDR-SB after one year. Only 198 participants

had both a usable image and a one-year outcome, and only 17 of them progressed (8.6%). Under five-fold cross-validation the fused model reached a mean AUROC of 0.786 (standard deviation 0.134) in the run reported at the review, with fold values between 0.55 and 0.96. A later run of the same notebook gave 0.652 (standard deviation 0.153), with fold values between 0.40 and 0.86. Tabular models on the same participants reached at most 0.78, again with standard deviations above 0.1.

When two runs of one design differ by more than 0.13 AUROC, no conclusion can be drawn from the result; with 17 events, a few participants determine the score. A review of the prototype also found technical problems. Slices of dynamic PET acquisitions were ordered by position only, so different time frames could be mixed in one volume; no spatial registration was applied; one cardiovascular variable was read from a column that does not exist in the release; and the SHAP analysis was fitted on all participants rather than on unseen ones. These problems led to the design used in this thesis: the full clinical cohort, three-year outcomes with thousands of events, and a separate, validated treatment of the PET images (Chapter 5).

1.4 Problem Statement

Let $\mathbf{x}_{t_0}$ be the information available about a person without dementia at a clinic visit t_0: demographics, cardiovascular and medical history, neuropsychological test results, the clinician's rating, genetics and, when available, PET measures. The problem is to estimate three quantities over the following three years:

$$\hat{p}_{\text{dem}} = P(\text{dementia diagnosed in } (t_0, t_0 + 3\,\text{y}] \mid \mathbf{x}_{t_0}), \tag{1.1}$$

$$\hat{p}_{\text{dec}} = P(\text{CDR-SB rises by} \geq 1 \text{ in } (t_0, t_0 + 3\,\text{y}] \mid \mathbf{x}_{t_0}), \tag{1.2}$$

$$\hat{\Delta} = E(\text{CDR-SB}_{t_0+3\,\text{y}} - \text{CDR-SB}_{t_0} \mid \mathbf{x}_{t_0}), \tag{1.3}$$

to determine how much each kind of information contributes, and to explain each estimate in terms a clinician can check, without using any information that would not have been available at t_0.

1.5 Research Questions and Objectives

The study answers seven research questions:

RQ1. How accurately can progression to dementia and cognitive decline within three years be predicted, and how does this differ between cognitively normal people and people with MCI?

RQ2. Does cardiovascular and medical information improve the prediction beyond demographics, genetics and current cognition?

RQ3. Do amyloid and tau PET measures improve the prediction?

RQ4. Do deep-learning models, including a modality-encoder fusion network, outperform classical models on this kind of data?

RQ5. Does information from earlier visits improve the prediction beyond the current visit?

RQ6. Can one model give calibrated estimates when whole kinds of information are missing?

RQ7. Can each estimate be explained and supported with published evidence in a form a clinician can read?

The objectives were to build a leakage-free cohort and outcomes from the NACC database; to harmonise neuropsychological tests across form versions; to train and compare classical and deep models under a person-level split with held-out centres; to measure the contribution of each kind of information by ablation; to explain predictions with Shapley values and attention weights; to test the added value of PET and of earlier visits; to validate the models in subgroups, for calibration and in sensitivity analyses; to build and validate a PET image pipeline; to deliver a clinician-facing prototype with a plain-language, evidence-backed report; and to verify the integrity of the entire pipeline with independent checks.

1.6 Contributions

- A reproducible pipeline that turns the NACC database into leakage-free three-year prediction tasks for dementia and for cognitive decline, with neuropsychological tests harmonised across form versions.
- A controlled comparison of classical models, deep tabular models and a modality-encoder fusion network, showing that model complexity brings no gain on these data.
- A measured answer to the cardiovascular question: once current cognition is known, cardiovascular and medical history adds about 0.001 AUROC for dementia and nothing measurable for CDR-SB decline.
- Results reported separately for cognitively normal participants and participants with MCI, with a quantification of how strongly the predictions depend on current cognitive status.
- A PET image pipeline whose amyloid measurements agree with the PET core's (correlation 0.98).

- A research prototype that checks the plausibility of its inputs and writes a plain-language report with quoted evidence.
- Seventeen independent verification checks, which found and corrected two sources of information leakage.

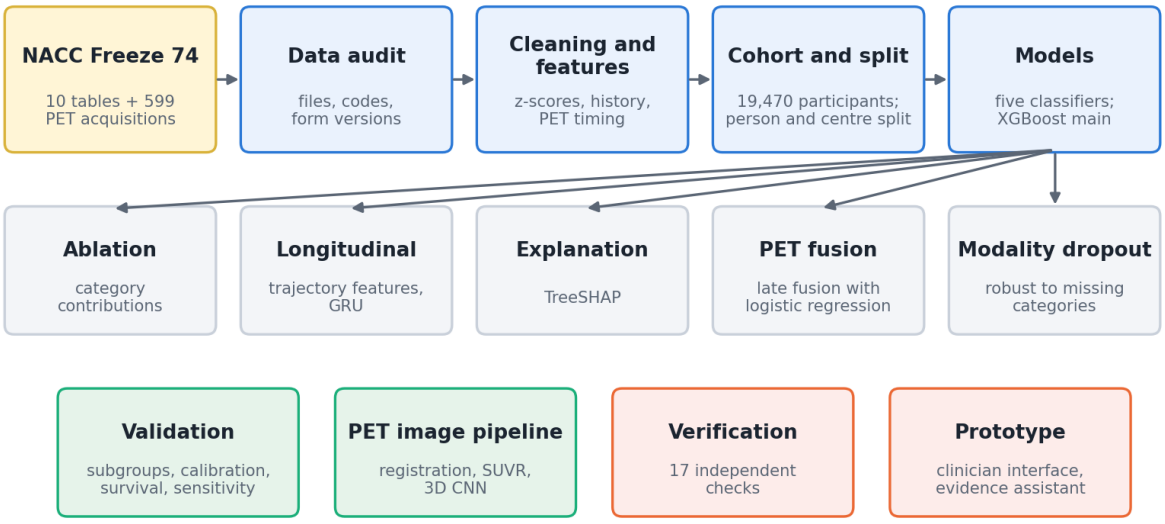


Figure 1.1: Overview of the work presented in this thesis.

1.7 Organisation of the Thesis

Chapter 2 reviews related work. Chapter 3 describes the data. Chapter 4 sets out the prediction tasks, cohort, features, models and evaluation. Chapter 5 describes the PET image pipeline. Chapter 6 reports the results. Chapter 7 describes the prototype. Chapter 8 explains how the pipeline was verified. Chapter 9 discusses the findings and their limits, and Chapter 10 concludes. The appendices list the features and the steps needed to reproduce the work.

Chapter 2

Literature Review

2.1 Dementia, Mild Cognitive Impairment and Their Definition

Clinical diagnosis of dementia rests on a decline in cognition that interferes with independence in daily activities, whereas MCI denotes measurable impairment with preserved independence. Alzheimer's disease, the most common cause, has increasingly been defined biologically rather than by its clinical presentation. The Alzheimer's Association workgroup updated the 2018 research framework in 2024, defining the disease by biomarkers of its characteristic pathology, amyloid plaques and tau tangles, measured in fluid or by PET, and staging it separately from the clinical syndrome [8]. This distinction matters for prediction studies: an outcome defined by clinical diagnosis, as in this thesis, measures progression of the syndrome, not of the underlying pathology.

2.2 Modifiable and Cardiovascular Risk Factors

The 2020 report of the Lancet Commission on dementia prevention described a life-course model with twelve potentially modifiable risk factors, including hypertension, diabetes, obesity, smoking and physical inactivity, which together account for around 40% of dementias worldwide [1]. Hypertension, particularly in midlife, has been related to a higher risk of cognitive decline and dementia, although the association is complex and differs with age at measurement [2]. Hypertension is the main risk factor for cerebral small vessel disease, a leading vascular contributor to cognitive impairment [3]. A recent meta-analysis found that higher cumulative systolic blood pressure was associated with incident dementia (odds ratio 1.21, 95% CI 1.00–1.45), while associations with diastolic pressure were inconsistent [9]. For diabetes, a meta-analysis of 15 studies reported a pooled relative risk of dementia of 1.59 (95% CI 1.40–1.80) [4], and among people with MCI, diabetes or the metabolic syndrome was associated with faster progression to dementia (pooled odds ratio 1.67, 95% CI 1.27–2.19) [10]. In people without dementia, a higher Framingham car-

diovascular risk score has been associated with more small vessel disease and with lower cerebrospinal fluid amyloid-β, with small vessel disease mediating the association [11].

These studies establish associations, mostly with exposure measured in midlife or accumulated over many years. They do not by themselves show how much cardiovascular information adds to a prediction model that already contains measurements of current cognition, which is the question addressed in this thesis.

2.3 Genetic Risk

The $\varepsilon 4$ allele of the APOE gene is the major genetic risk factor for late-onset Alzheimer's disease; APOE is the main cholesterol carrier in the brain and is involved in lipid transport and synaptic function [5]. In cohort data the genotype is not always available, and whether a participant was genotyped is not random, which has to be handled explicitly when building features.

2.4 Amyloid and Tau PET

PET with amyloid tracers measures cortical amyloid deposition, tau tracers measure neurofibrillary tangle pathology, and FDG measures glucose metabolism as an index of neuronal activity [6]. Tau PET is technically more demanding than amyloid PET, with tracer-specific binding properties and off-target signal [7]. Because amyloid SUVR values depend on the tracer and the processing pipeline, the Centiloid scale was introduced to express amyloid burden on a common scale anchored at young controls (0) and typical Alzheimer's dementia (100) [12]. These properties motivate the use of Centiloids for amyloid and within-tracer standardisation for tau in this thesis.

2.5 Machine Learning for Tabular Clinical Data

Clinical cohort data are tabular: a mixture of continuous, ordinal and categorical variables with substantial missingness. Logistic regression remains the reference method in clinical prediction because it is transparent and often well calibrated. Ensembles of decision trees, such as random forests [13] and gradient-boosted trees as implemented in XGBoost [14], capture non-linear effects and interactions and handle missing values naturally. Deep-learning models for tables have been proposed repeatedly; in a systematic comparison, Gorishniy et al. introduced the FT-Transformer, which embeds each feature as a token and applies self-attention, and found it to be a strong deep baseline while also observing that no deep model was universally superior to gradient-boosted trees [15]. For sequences of visits, recurrent networks such as the gated recurrent unit [16] can read a participant's history directly.

Evaluation of prediction models requires more than discrimination. Calibration, clinical usefulness assessed by decision curve analysis [17], and robustness to censoring, handled with survival methods such as the Kaplan–Meier estimator [18] and the Cox model [19], are all needed before a model can be interpreted.

2.6 Explainability

Shapley additive explanations (SHAP) attribute a prediction to the input features using Shapley values from cooperative game theory, with the property that the contributions sum to the prediction [20]. For tree ensembles, exact values can be computed efficiently with TreeSHAP [21]. For convolutional networks, gradient-weighted class activation mapping (Grad-CAM) highlights image regions that influence the output [22]. In both cases the explanation describes what the model uses, not what causes the outcome.

2.7 Retrieval-Augmented Generation

Language models can produce fluent but unsupported statements, which is a particular risk in medicine. Retrieval-augmented generation conditions the generated text on passages retrieved from a defined collection [23]. Retrieval can use sentence embeddings [24], keyword ranking functions such as BM25 [25], or a combination of rankings, for which reciprocal rank fusion is a simple and effective method [26].

2.8 Related Prediction Studies

Several studies have combined cardiovascular, clinical and genetic information to predict dementia. In the NACC database, Alaka et al. added resting heart rate to the CAIDE cardiovascular dementia risk score using a random forest and reported improved discrimination in most racial and ethnic groups (mean AUC 0.80–0.91), but not in American Indian participants [27]. A systematic review of 22 studies found that 14 of 15 cardiovascular risk models were associated with dementia or cognitive decline, but that comparisons between models were hampered by differences in cohorts, tests and designs [28]. In the Rush Memory and Aging Project, a higher Framingham cardiovascular risk score was associated with faster decline in global cognition and memory and with smaller hippocampal volume [29].

Among machine-learning studies, Stern and Linial predicted Alzheimer's disease in the UK Biobank from diagnosis codes, APOE genotype and modifiable risk factors, with CatBoost performing best (ROC-AUC 0.773) [30]. Aschwanden et al. ranked 52 predictors of cognitive impairment and dementia in the Health and Retirement Study with random survival forests and Cox models, and found that cardiovascular factors and polygenic scores were less important than expected when evaluated together with sociodemographic,

psychological and health variables [31]. For diagnosis rather than prediction, Yi et al. classified normal cognition, MCI and Alzheimer's disease with a class-weighted XGBoost model explained by SHAP, reporting AUCs of 0.91 on ADNI and 0.88 on NACC [32], and El-Sappagh et al. combined eleven modalities from ADNI in a two-layer random forest model for diagnosis and progression detection with explanations for each decision [33].

These studies show that cardiovascular information is associated with cognitive outcomes and that tree ensembles with SHAP are effective for these data. Fewer studies measure the added value of cardiovascular information after current cognitive status is accounted for, and few report results separately for cognitively normal participants and participants with MCI.

2.9 Data Source

The National Alzheimer's Coordinating Center maintains the Uniform Data Set, a standardised longitudinal assessment administered at Alzheimer's Disease Research Centers across the United States. Version 3 of the neuropsychological battery, introduced in 2015, replaced several proprietary tests with new ones [34], which creates the harmonisation problem addressed in Chapter 4.

2.10 Research Gap

The epidemiological literature establishes that vascular risk factors, APOE and amyloid and tau pathology are associated with dementia. Prediction studies frequently report high discrimination, but such figures depend strongly on who is included and on whether current cognitive status is among the inputs. Three questions are less often answered together on one large cohort under a strict, leakage-controlled design: how much each source of information adds once current cognition is known, whether deep-learning models improve on classical ones for this kind of data, and how results differ between cognitively normal participants and participants with MCI. This thesis addresses these questions on the NACC database, reports results by baseline diagnosis, and accompanies the model with a validated PET image pipeline, a clinician-facing prototype and independent verification of the pipeline.

Chapter 3

Data

3.1 The National Alzheimer's Coordinating Center

The study uses the Freeze 74 release (June 2026) of the National Alzheimer's Coordinating Center (NACC) database. NACC pools longitudinal data from Alzheimer's Disease Research Centers across the United States. Volunteers are assessed roughly once a year with the Uniform Data Set (UDS), a fixed battery of questionnaires, examinations and neuropsychological tests that every centre administers in the same way [34]. The release used here contains 217,598 visits from 57,038 participants at 46 centres, collected between June 2005 and May 2026. Participants are referred through memory clinics or volunteer directly, so the cohort is a research sample and not a population sample: at the first visit the median age is 71 years, median education is 16 years and 76% of participants identify as White.

Access to the data is governed by the NACC data use agreement. All analysis was performed on a local machine. No participant-level value, identifier or image file name appears in this thesis, in the code repository or in any shared report; every reported number is an aggregate.

3.2 Files received

Table 3.1 lists the ten tables in the release. Only the main clinical table and the four summary PET tables contribute model inputs; the remaining files were audited but do not supply features.

Table 3.1: Files in the NACC release and their role in this study.

File	Rows	Participants	Role
Main clinical table (`ftldlbd`)	217,598	57,038	All clinical features and the outcome
SCAN amyloid PET summary	3,933	3,566	Centiloids and amyloid status
Mixed-protocol amyloid PET summary	2,235	2,073	Same, for older scans
SCAN tau PET	2,653	2,405	Meta-temporal and entorhinal SUVR
Mixed-protocol tau PET	671	616	Same, for older scans
SCAN and mixed-protocol amyloid PET, regional	6,168	—	Not used (tracer-specific SUVR)
SCAN FDG PET	885	639	Not used (about 600 linked participants)
SCAN PET quality-control list	8,759	4,585	Audit of the local image folders
CLARiTI study tracking	857	776	Not used (administrative)

In the clinical table one row is one visit; in the PET tables one row is one scan. The participant identifier is the only key shared by all tables. Because a scan is not tied to a visit number, each scan has to be matched to a visit by date (Section 4.3.4).

In addition, 599 raw PET acquisitions (318 amyloid, 232 tau and 49 FDG scans from 446 participants, about 100 GB of DICOM files) were available locally. They are used in Chapter 5.

3.3 The clinical table

The clinical table has 2,649 columns, but most of them are empty for most visits: modules for frontotemporal and Lewy body disease, autopsy findings, form version 4 items and free text. Only 787 columns contain data in at least 10% of visits. From these, 114 variables were mapped by hand to a meaning, a category, a role and their own missing-value codes, following the NACC Researcher's Data Dictionary. Seventy-six of them are used to build the model inputs.

3.3.1 Missing-value codes

NACC encodes non-responses with numeric codes whose meaning depends on the variable: -4 marks a question that was not part of the visit, codes such as 8, 88 or 888 mark "not applicable", and 9, 99 or 999 mark "unknown". These codes cannot be removed globally. The value 88, for example, is a genuine age in 3,655 visits and a genuine diastolic pressure in 3,090 visits, but means "never smoked" in the years-smoked column (73,669 visits). Cleaning was therefore performed column by column from the variable map.

3.3.2 Changes in the forms

The UDS has been revised three times. The most important change for this study took place with version 3 in 2015, when the neuropsychological battery was replaced: the MMSE, Logical Memory, Digit Span and Boston Naming tests were replaced by the MoCA, Craft Story, Number Span and MINT tests. Only category fluency and the Trail Making tests are common to all versions. The same participant can therefore have different tests at different visits. Health history is also recorded on two forms: the participant-reported history (form A5) and, in version 3, the clinician-assessed conditions (form D2). Telephone visits contain no neuropsychological tests. Figure 3.1 shows how the availability of each test changes between versions; it is the reason the tests were converted into common domain scores (Section 4.3.1).

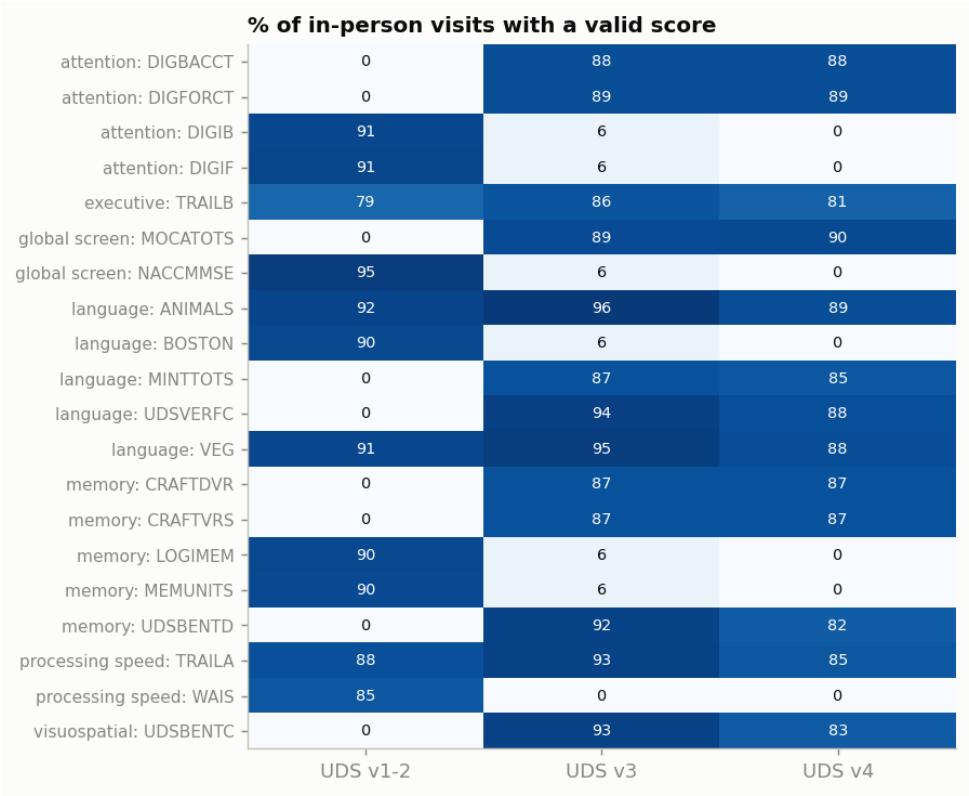


Figure 3.1: Percentage of in-person visits with a valid score for each neuropsychological test, by UDS version. Tests of versions 1–2 and of version 3 are almost disjoint.

3.4 Feature categories

The variables used are grouped into five categories (Table 3.2). The complete list, with source columns and value codes, is given in Appendix A.

Table 3.2: Feature categories, source columns and number of model features.

Category	Main source columns	Features
Demographics	NACCAGE, NACCSEX, EDUC, NACCNIHR, NACCHISP, NACCLIVS	8
Cardiovascular and medical	Twelve conditions from forms A5 and D2, smoking, blood pressure, heart rate, BMI, four medication classes and the medication count	25
Cognitive tests	Twenty neuropsychological tests, the Geriatric Depression Scale and twelve NPI-Q behaviour items	9
Clinical stage	CDRSUM, NACCUDSD	3
Genetics	NACCNE4S, NACCFAM	3
PET (second-stage model only)	CENTILOIDS, amyloid status, meta-temporal and entorhinal tau SUVR	4

Four groups of variables were deliberately excluded. Autopsy findings are known only after death and would leak future information. The Functional Activities Questionnaire, the level of independence, subjective and informant-reported memory complaints and the use of Alzheimer's medication are so close to the diagnosis itself that including them would partly reproduce the clinician's decision; they were evaluated separately as a sensitivity analysis. Regional PET values were not used because they are tracer-specific and highly redundant. Form version 4 visits (3% of visits) were excluded because they use new variable names.

3.5 PET data

Amyloid PET values are available for about 5,100 participants who also appear in the clinical table, and tau PET values for 2,724. The amyloid tables provide the Centiloid value, which is comparable across tracers, and a positive or negative status assigned by the PET core. The tau tables provide standardised uptake value ratios (SUVR) in about 160 regions; the meta-temporal composite and the entorhinal cortex were retained as the standard summaries of early tau deposition. Because the two tau tracers (flortaucipir and MK-6240) give different SUVR scales, each tau value was standardised within tracer against amyloid-negative participants from the training set. Only scans that passed quality control were used.

Most PET scans are recent, so few scanned participants have three years of clinical follow-up after the scan. This limits what can be learned from PET in this cohort (Section 6.7).

Chapter 4

Methodology

4.1 Prediction task

The prediction is made at an *index visit*, using only information available at or before that visit. For a participant without dementia at the index visit, the task is to estimate the probability of a clinical diagnosis of dementia within three years:

$$\hat{p} = P\left(\text{dementia diagnosed in } (t_0, t_0 + 3 \text{ years}] \mid \mathbf{x}_{t_0}\right), \tag{4.1}$$

where t_0 is the date of the index visit and $\mathbf{x}_{t_0}$ the feature vector at that visit. The main analysis uses each participant's first visit as the index visit. If T denotes the time from t_0 to the first visit with a dementia diagnosis and C the time to the last visit, the binary label is

$$y = \begin{cases} 1 & \text{if } T \leq 3, \\ 0 & \text{if } T > 3 \text{ and } C \geq 3, \\ \text{unknown} & \text{if } T > 3 \text{ and } C < 3, \end{cases} \tag{4.2}$$

with times in years.

4.1.1 Choice of horizon and outcome

Clinical diagnosis (`NACCUDSD`) and the Clinical Dementia Rating are recorded at every visit in every form version, which makes them the most consistent outcomes available. A dementia diagnosis was chosen as the main outcome because it is clinically meaningful and easy to interpret. The three-year horizon balances the number of events against loss to follow-up: Kaplan–Meier estimates [18] show that about 35% of participants with MCI progress to dementia within three years, and that 11% of cognitively normal participants progress to MCI or dementia, while the number of participants with a known status falls steadily with longer horizons and most PET scans are too recent for a five-year horizon. The Kaplan–Meier estimate of the probability of remaining free of the event up to time t

is

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right), \tag{4.3}$$

where d_i is the number of events at time t_i and n_i the number of participants still under observation just before t_i; the three-year progression rate is $1 - \hat{S}(3)$ (Figure 4.1).

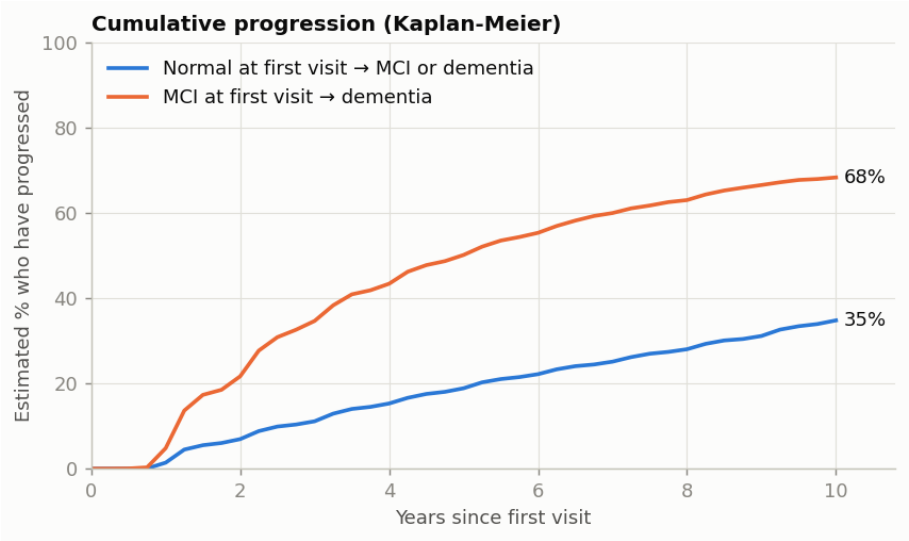


Figure 4.1: Kaplan–Meier estimates of cumulative progression from the first visit: cognitively normal participants to MCI or dementia, and participants with MCI to dementia.

4.1.2 Censoring

A participant last seen less than three years after the index visit without a dementia diagnosis has an unknown outcome. Such participants were excluded from the binary task rather than counted as negative, which would bias the estimated risk downwards. Because this exclusion could itself introduce bias, a Cox proportional hazards model [19] that uses all eligible participants, including the censored ones, was fitted as a robustness check (Section 6.10). Figure 4.2 illustrates the three possible cases.

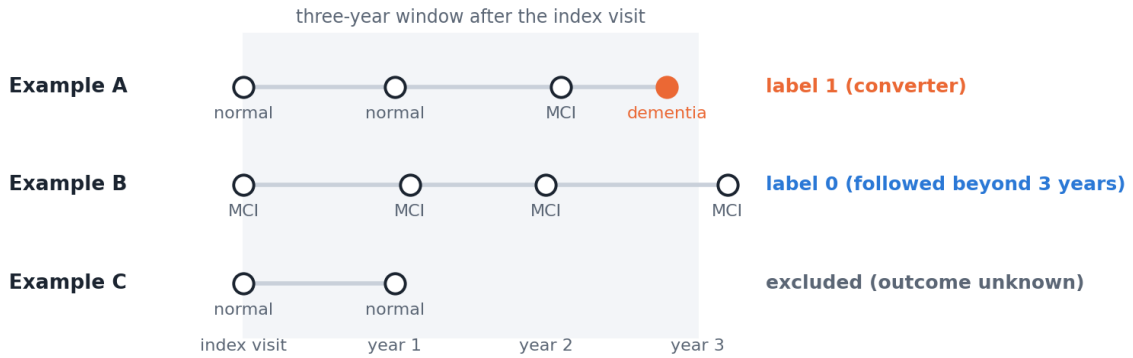


Figure 4.2: Definition of the outcome. Only the index visit supplies features; later visits supply the label.

4.1.3 Secondary outcomes: cognitive decline

Dementia is one end point of cognitive decline, but many people worsen without reaching it. Two further outcomes were therefore defined from the CDR Sum of Boxes, which is recorded at every visit in every form version. The binary decline outcome is a rise of at least one point in CDR-SB at any visit within three years of the index visit; it is unknown for participants who neither rose by a point nor were followed for three years. The continuous progression outcome is the change in CDR-SB between the index visit and the visit closest to three years later, accepted if that visit falls within six months of the three-year mark. Writing $c(t)$ for CDR-SB at time t after the index visit and v_i for the visit times,

$$y_{\text{dec}} = \mathbb{1}\Big[\max_{0<v_i\le 3} c(v_i) - c(0) \ge 1\Big], \qquad \Delta = c(v^*) - c(0), \quad v^* = \operatorname*{arg\,min}_{|v_i-3|\le 0.5} |v_i - 3|. \tag{4.4}$$

The same eligibility rules and splits are used for all outcomes.

4.2 Cohort

Table 4.1 shows how the cohort was formed. Visits had to be in person, because telephone visits lack the neuropsychological tests, and on form versions 1 to 3, because version 4 reorganised many of the other variables and accounts for only 3% of visits. Participants with dementia at the index visit, or at any earlier visit, are not eligible.

Table 4.1: Cohort flow.

Step	Participants
All participants	57,038
First visit in person, form version 1 to 3	55,461
and not demented at that visit	38,107
and outcome known at three years	19,470
of whom diagnosed with dementia within three years	2,818 (14.5%)

The "not demented" cohort contains three clinically different groups: 12,377 cognitively normal participants (198 converters, 1.6%), 5,993 participants with MCI (2,485 converters, 41.5%) and 1,100 participants classed as impaired but not MCI (135 converters, 12.3%). Results are reported for the whole cohort and for the normal and MCI groups separately.

4.3 Feature construction

4.3.1 Harmonised cognitive scores

To place tests from different form versions on one scale, each raw score s was converted into a demographically adjusted z-score. For each test, a linear model of the score on age, sex and education was fitted among cognitively normal participants at their first visit, using the training set only:

$$\hat{s} = \beta_0 + \beta_1\,\text{age} + \beta_2\,\text{female} + \beta_3\,\text{education}, \qquad z = \frac{s - \hat{s}}{\sigma_{\text{res}}}, \tag{4.5}$$

where σ_{res} is the residual standard deviation of the fitted model. The sign is reversed for the two timed Trail Making tests, so that a negative z always means worse than expected. Values were clipped to $[-5, 5]$. The z-scores of the tests in one cognitive domain were then averaged to give six domain scores: global screening (MoCA, MMSE), memory, attention, speed and executive function, language, and visuospatial function. A participant receives a domain score from whichever tests their visit contained.

As an example, for the MoCA the fitted model is $\hat{s} = 23.0 - 0.059\,\text{age} + 0.33\,\text{female} + 0.42\,\text{education}$ with $\sigma_{\text{res}} = 2.56$, estimated from 6,264 normal participants. A 72-year-old woman with 16 years of education has an expected score of 25.8, so a score of 21 corresponds to $z = -1.9$ (Figure 4.3).

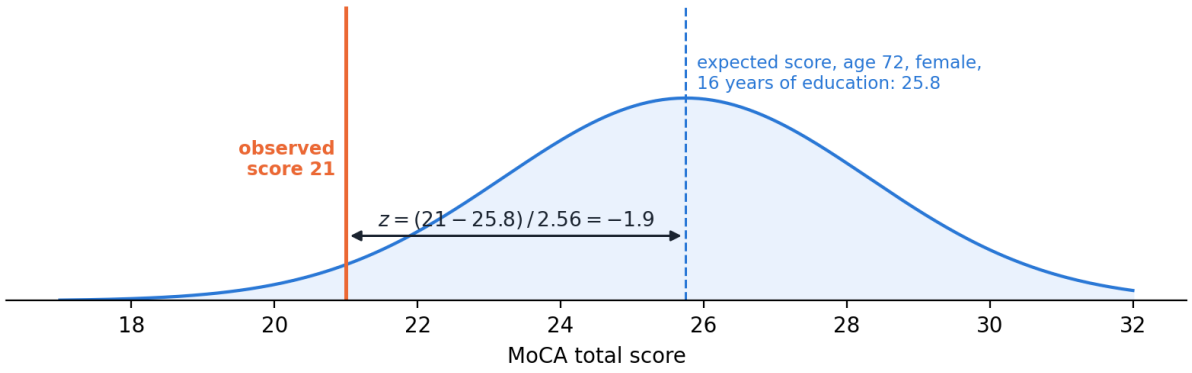


Figure 4.3: Demographically adjusted z-score for the MoCA, using the fitted normative model.

The harmonisation was checked on cognitively normal participants who were not used to fit the norms: their mean domain z-scores lie within 0.1 of zero under every form version, while participants with MCI and dementia score progressively lower (Figure 4.4). The scores therefore mean the same thing whichever tests a visit contained.

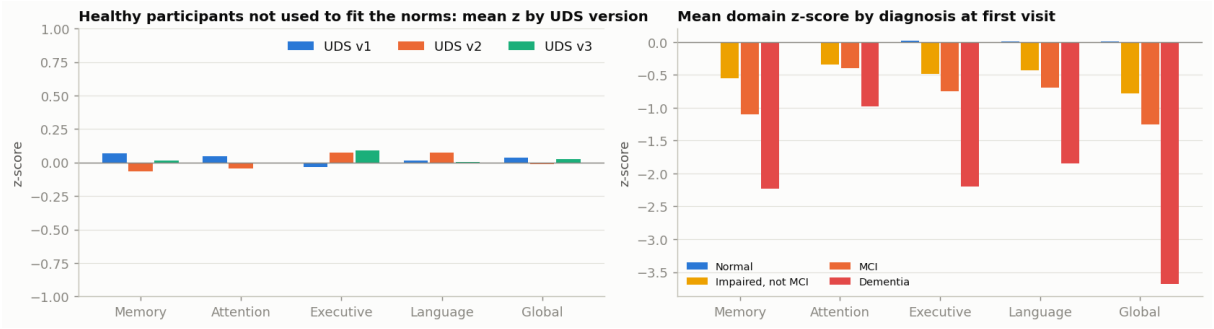


Figure 4.4: Left: mean domain z-score of cognitively normal participants not used to fit the norms, by UDS version. Right: mean domain z-score by diagnosis at the first visit.

Three further cognitive features were derived: the number of tests the participant could not complete for cognitive or behavioural reasons (NACC code 96 or 996), which is informative about severity and is therefore kept rather than treated as missing; the Geriatric Depression Scale total; and the number of NPI-Q behavioural symptoms present.

4.3.2 Cardiovascular history

Each condition k was coded as "ever present", $h_k(t) = \max_{v_i \le t} r_k(v_i)$, where $r_k(v_i) = 1$ if the condition is recorded at visit v_i. The record was taken from the participant-reported history (form A5, where both recent and remote mentions count) and, where that form was blank, from the clinician-assessed form D2. Once present, a condition remains present at all later visits. Measured blood pressure, heart rate and BMI use the most recent measurement up to the visit. Two summary features were derived: pulse pressure, $\mathrm{PP} = \mathrm{BP}_{\mathrm{sys}} - \mathrm{BP}_{\mathrm{dia}}$, and the vascular risk count, $R = h_{\mathrm{HTN}} + h_{\mathrm{chol}} + h_{\mathrm{DM}} + h_{\mathrm{smoke}} \in \{0, \dots, 4\}$.

4.3.3 Genetics

The number of APOE $\varepsilon 4$ alleles is used directly. About one participant in five was not genotyped; genotyping depends on centre and consent and is therefore not random, so these participants receive a separate "not genotyped" indicator instead of an imputed value.

4.3.4 Preventing the use of future information

No feature at a visit may depend on a later visit. Fixed characteristics such as education are carried forward from the visit where they were first recorded, never backwards. A PET scan is used at a visit only if it was acquired on or up to 365 days before that visit. Both rules were introduced after the verification step described in Chapter 8 found violations in an earlier version of the pipeline.

4.4 Data splits

All splits were made at participant level, so that all visits of a person fall in the same partition. Twenty percent of the centres (9 of 46) were held out entirely to measure performance at sites the model has never seen. The remaining participants were divided into training (70%), validation (15%) and test (15%) sets, stratified by diagnosis at the first visit. The random seed was fixed (2026). Table 4.2 gives the resulting sizes for the first-visit cohort.

Table 4.2: Partition sizes for the main cohort.

Partition	Participants	Converters
Training	10,879	1,599
Validation	2,241	342
Test	2,286	332
Held-out centres	4,064	545

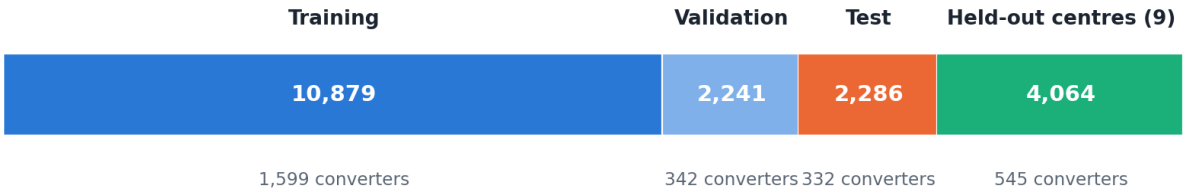


Figure 4.5: Participant-level partitions of the main cohort.

Every quantity estimated from data (normative models for z-scores, tau standardisation, imputation medians, scaling) was fitted on the training set and applied unchanged to the other sets. The validation set was used for early stopping and model selection; the test set and the held-out centres were used only for evaluation.

4.5 Models

Five classifiers of increasing complexity were compared on the same 48 features (Table 4.3); their mathematical form is given in Section 4.5.1. Logistic regression is the standard baseline in clinical prediction. Random forests [13] and gradient-boosted trees (XGBoost [14]) are strong learners for tabular data that handle non-linear effects and interactions. A multilayer perceptron and the FT-Transformer [15] represent deep learning; the latter tokenises each feature and applies self-attention, and is among the strongest deep architectures reported for tabular data. Except for logistic regression, every model was trained with three random seeds and the predictions were averaged.

Table 4.3: Models and settings.

Model	Settings
Logistic regression	Median imputation with missing-value indicators, standardisation; regularisation strength chosen from five values on the validation set
Random forest	500 trees, minimum 10 samples per leaf, $\sqrt{p}$ features per split
XGBoost	Up to 3,000 trees, learning rate 0.03, depth chosen from {3, 4, 6} on validation AUROC, subsample and column sample 0.8, early stopping after 100 rounds without improvement; missing values handled natively
Multilayer perceptron	Two hidden layers (128 and 64 units), dropout 0.2, AdamW, early stopping on validation AUROC
FT-Transformer	Feature tokens of dimension 64, three Transformer layers with eight heads, a learned token for each missing feature, early stopping on validation AUROC

XGBoost was used as the main model for all further experiments because it performed as well as the others, handles missing values without imputation and allows exact Shapley value explanations. For the progression outcome, the same gradient-boosting settings were used with an absolute-error objective, and the model was compared with the simplest reference, predicting no change.

4.5.1 Mathematical form of the models

Let $\mathbf{x} \in \mathbb{R}^p$ be the feature vector of a participant, $y \in \{0, 1\}$ the label and $\sigma(u) = 1/(1 + e^{-u})$ the logistic function. All classifiers output a probability $\hat{p} = \sigma(f(\mathbf{x}))$ and, apart from the random forest, are fitted by minimising the binary cross-entropy over the n training participants,

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \Big], \tag{4.6}$$

plus a model-specific penalty.

Logistic regression. After median imputation, missing-value indicators and standardisation give a vector $\tilde{\mathbf{x}}$, the model is $f(\mathbf{x}) = \beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}$ with an L_2 penalty,

$$\min_{\beta_0, \boldsymbol{\beta}} \ \frac{1}{2} \|\boldsymbol{\beta}\|_2^2 + C \sum_{i=1}^{n} \ell(y_i, \hat{p}_i), \qquad C \in \{0.01, 0.03, 0.1, 0.3, 1\}, \tag{4.7}$$

where ℓ is the per-participant cross-entropy and C was chosen on the validation set.

Random forest. The forest averages the class-probability estimates of $B = 500$ decision trees, each grown on a bootstrap sample with $\lfloor \sqrt{p} \rfloor$ candidate features per split: $\hat{p}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \hat{p}_b(\mathbf{x})$.

Gradient-boosted trees (XGBoost). The score is a sum of regression trees, built one at a time:

$$f_M(\mathbf{x}) = \sum_{m=1}^{M} \eta\, g_m(\mathbf{x}), \qquad \eta = 0.03. \tag{4.8}$$

Tree g_m is chosen to minimise a second-order approximation of the regularised objective

$$\sum_{i=1}^{n} \left[G_i\, g_m(\mathbf{x}_i) + \tfrac{1}{2} H_i\, g_m(\mathbf{x}_i)^2 \right] + \gamma J + \tfrac{1}{2}\lambda \sum_{j=1}^{J} w_j^2, \tag{4.9}$$

where $G_i = \hat{p}_i - y_i$ and $H_i = \hat{p}_i(1-\hat{p}_i)$ are the first and second derivatives of the loss at the current model, J is the number of leaves, w_j the leaf values, $\lambda = 2$ and $\gamma = 0$. For a leaf containing the participant set I_j the optimal value is $w_j^* = -\sum_{i\in I_j} G_i / (\sum_{i\in I_j} H_i + \lambda)$, and a split is kept only if every child has $\sum H_i \geq 5$ (minimum child weight). Missing values are sent down a default branch learned from the data. For the progression outcome the loss is the absolute error $|\Delta_i - f(\mathbf{x}_i)|$, whose minimiser is the conditional median.

Multilayer perceptron. With $\tilde{\mathbf{x}}$ as for logistic regression,

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\tilde{\mathbf{x}} + \mathbf{b}_1), \quad \mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2), \quad f(\mathbf{x}) = \mathbf{w}_3^\top \mathbf{h}_2 + b_3, \tag{4.10}$$

with 128 and 64 hidden units and dropout of 0.2 after each hidden layer.

FT-Transformer. Each feature j becomes a token $\mathbf{e}_j = x_j \mathbf{w}_j + \mathbf{b}_j \in \mathbb{R}^{64}$, or a learned token $\mathbf{m}_j$ when x_j is missing. A learned classification token $\mathbf{e}_0$ is prepended, and three pre-norm Transformer layers are applied. Each layer computes multi-head self-attention,

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\!\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{4.11}$$

with eight heads ($d_k = 8$), followed by a feed-forward block with GELU activation; both have residual connections. The output for $\mathbf{e}_0$ passes through layer normalisation and a linear layer to give $f(\mathbf{x})$.

Both networks were trained with AdamW (learning rate 10^{-3} for the perceptron and 5×10^{-4} for the FT-Transformer, weight decay 10^{-4}) on batches of 256, and training stopped when the validation AUROC had not improved for eight epochs.

4.6 Modality-Encoder Fusion Network

To test the fusion architecture proposed at the start of the project, a neural network was built in which each of the five kinds of information has its own encoder. Each encoder is a two-layer network that maps the features of one modality to a 32-dimensional token. When a modality is entirely missing for a participant, its token is replaced by a learned

"missing" token, so the network can tell an absent modality from a normal one. A learned type embedding identifies each modality. A four-head attention layer, queried by a learned summary vector, combines the five tokens into one representation (Tier 1 fusion), and the attention weights show how much the network drew on each modality. Two heads share this representation: a risk head for dementia within three years and a progression head for the three-year change in CDR-SB, trained jointly with a binary cross-entropy loss and a Huber loss weighted by 0.5 on the participants with a known change. Modality dropout with probability 0.35 was applied during training. The network was trained with AdamW (learning rate 10^{-3}), early stopping on validation AUROC, and three seeds whose predictions were averaged. PET measures are added afterwards by the late-fusion model (Tier 2), because too few scanned participants have three-year outcomes to train them inside the network.

Formally, let $\mathbf{x}^{(g)}$ be the features of modality $g \in \{1, \dots, 5\}$ and $\pi_g \in \{0, 1\}$ indicate whether the modality is present. The token of modality g is

$$\mathbf{t}_g = \pi_g \, E_g\big(\mathbf{x}^{(g)}\big) + (1 - \pi_g)\, \mathbf{m}_g + \boldsymbol{\tau}_g, \qquad E_g(\mathbf{x}) = \mathbf{W}_{g2}\ \mathrm{ReLU}(\mathbf{W}_{g1}\mathbf{x} + \mathbf{b}_{g1}) + \mathbf{b}_{g2}, \tag{4.12}$$

where $\mathbf{m}_g$ is the learned missing token and $\boldsymbol{\tau}_g$ the type embedding, all in $\mathbb{R}^{32}$. For each attention head h, the learned query $\mathbf{q}$ gives the weights

$$\alpha_g^{(h)} = \frac{\exp\big((\mathbf{W}_Q^{(h)}\mathbf{q})^\top \mathbf{W}_K^{(h)}\mathbf{t}_g/\sqrt{8}\big)}{\sum_{g'} \exp\big((\mathbf{W}_Q^{(h)}\mathbf{q})^\top \mathbf{W}_K^{(h)}\mathbf{t}_{g'}/\sqrt{8}\big)}, \qquad \mathbf{z} = \mathrm{LN}\left(\mathbf{W}_O\big[\textstyle\sum_g \alpha_g^{(h)} \mathbf{W}_V^{(h)}\mathbf{t}_g\big]_{h=1}^{4}\right), \tag{4.13}$$

and the reported attention share of modality g is the average of $\alpha_g^{(h)}$ over the four heads. The risk head gives $\hat{p} = \sigma(r(\mathbf{z}))$ and the progression head $\hat{\Delta} = q(\mathbf{z})$, each a two-layer network. The training loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + 0.5\, \frac{1}{|\mathcal{D}_\Delta|} \sum_{i \in \mathcal{D}_\Delta} \rho\big(\Delta_i - \hat{\Delta}_i\big), \qquad \rho(u) = \begin{cases} \tfrac{1}{2}u^2 & |u| \le 1, \\ |u| - \tfrac{1}{2} & |u| > 1, \end{cases} \tag{4.14}$$

where $\mathcal{D}_\Delta$ is the set of participants in the batch whose three-year change is known and ρ is the Huber loss. Gradients were clipped to a norm of 1.

4.7 Contribution of each feature category

The contribution of a category was measured by ablation: the model was trained with all categories and again without the category of interest, and the paired difference in AUROC on the same participants was computed with bootstrap confidence intervals,

$$\Delta\mathrm{AUROC}_{\mathcal{G}} = \mathrm{AUROC}\big(f_{\mathrm{all}}\big) - \mathrm{AUROC}\big(f_{\mathrm{all}\setminus\mathcal{G}}\big). \tag{4.15}$$

Each category was also evaluated alone and in cumulative order.

4.8 Longitudinal models

To test whether earlier visits add information, the index visit was moved to a later "landmark" visit preceded by at least two earlier visits. Three models were compared on the same landmarks: XGBoost on the landmark visit only; XGBoost with slope and change features of the cognitive scores and CDR-SB over the earlier visits; and a gated recurrent unit (GRU) [16] with 48 hidden units that reads up to six visits, with the time of each visit as an input because visit intervals are irregular. For visit k with input $\mathbf{s}_k$ (standardised scores, missing-value flags and years before the landmark), the GRU updates its state as

$$\begin{aligned} \mathbf{r}_k &= \sigma(\mathbf{W}_r \mathbf{s}_k + \mathbf{U}_r \mathbf{h}_{k-1} + \mathbf{b}_r), & \mathbf{u}_k &= \sigma(\mathbf{W}_u \mathbf{s}_k + \mathbf{U}_u \mathbf{h}_{k-1} + \mathbf{b}_u), \\ \tilde{\mathbf{h}}_k &= \tanh\big(\mathbf{W}_h \mathbf{s}_k + \mathbf{U}_h(\mathbf{r}_k \odot \mathbf{h}_{k-1}) + \mathbf{b}_h\big), & \mathbf{h}_k &= (1 - \mathbf{u}_k) \odot \tilde{\mathbf{h}}_k + \mathbf{u}_k \odot \mathbf{h}_{k-1}, \end{aligned} \tag{4.16}$$

and the final state is joined with a 96-unit encoding of the landmark-visit features before a small output network. The trajectory features for XGBoost are, for each score, the least-squares slope over the visits up to the landmark, $\hat{b} = \sum_k (\tau_k - \bar{\tau})(s_k - \bar{s}) / \sum_k (\tau_k - \bar{\tau})^2$ with visit times τ_k, and the change since the previous visit.

4.9 PET late fusion

Because few scanned participants have three years of follow-up, a large model cannot be fitted on PET participants alone. A late-fusion design was used: the clinical model, trained on all participants, produces a risk $\hat{p}_{\text{clin}}$ for each scanned participant; a small, strongly regularised logistic regression then combines the logit of that risk with the PET measures $\mathbf{u}$ and indicators $\boldsymbol{\delta}$ of which measures are present:

$$\operatorname{logit} \hat{p}_{\text{PET}} = \gamma_0 + \gamma_1 \log \frac{\hat{p}_{\text{clin}}}{1 - \hat{p}_{\text{clin}}} + \boldsymbol{\gamma}_u^{\top} \mathbf{u} + \boldsymbol{\gamma}_\delta^{\top} \boldsymbol{\delta}. \tag{4.17}$$

Because few events are available, the L_2 penalty was set firmly ($C = 0.3$ in the final model). Two rules for matching scans to visits were compared: a strict rule (scan on or before the index visit) and a nearest-visit rule (larger, but the scan may follow the visit by a few weeks).

4.10 Modality dropout

In practice not every patient has every assessment. A model trained on complete records may behave poorly when an entire category is missing. To make one model usable with any combination of available information, the training data were augmented with four additional copies in which each optional category (genetics, cardiovascular, cognitive tests, clinical stage) was removed independently with probability 0.35. For copy k of participant

i and optional category g,

$$\tilde{\mathbf{x}}_{ik}^{(g)} = \begin{cases} \text{missing} & \text{if } \xi_{ikg} = 1, \\ \mathbf{x}_i^{(g)} & \text{otherwise,} \end{cases} \qquad \xi_{ikg} \sim \text{Bernoulli}(0.35), \tag{4.18}$$

so that the model sees every pattern of missing categories during training, while demographics are always kept. This model was compared with a standard model and with specialist models retrained for each pattern of missing categories.

4.11 Explainability

Predictions were explained with Shapley additive explanations (SHAP) [20], computed exactly for the tree ensemble with TreeSHAP [21]. For each test participant, SHAP decomposes the predicted log-odds into a sum of feature contributions,

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j, \qquad \phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x}) \Big], \tag{4.19}$$

where F is the set of all features, f_S the expected model output when only the features in S are known, and ϕ_0 the average output over the training data. Global importance is the mean absolute contribution of a feature over the test set, $I_j = \frac{1}{n} \sum_i |\phi_{ij}|$; the importance of a category $\mathcal{G}$ is $\sum_{j \in \mathcal{G}} I_j$, reported as a share of $\sum_j I_j$. Stability was assessed by refitting the model on eight bootstrap samples of the training set.

4.12 Evaluation

Discrimination was measured by the area under the ROC curve (AUROC) and the area under the precision–recall curve (AUPRC). The AUROC equals the probability that a randomly chosen converter receives a higher predicted risk than a randomly chosen non-converter,

$$\text{AUROC} = \frac{1}{n_1 n_0} \sum_{i:y_i=1} \sum_{j:y_j=0} \Big(\mathbb{1}[\hat{p}_i > \hat{p}_j] + \tfrac{1}{2} \mathbb{1}[\hat{p}_i = \hat{p}_j] \Big), \tag{4.20}$$

where n_1 and n_0 are the numbers of converters and non-converters. The AUPRC was computed as average precision, $\sum_k (R_k - R_{k-1}) P_k$, over the thresholds k with precision P_k and recall R_k; for an uninformative model it equals the event rate.

Calibration was assessed by the Brier score, the calibration slope and intercept, and by comparing predicted and observed risk in deciles:

$$B = \frac{1}{n} \sum_{i=1}^{n} (\hat{p}_i - y_i)^2, \qquad \operatorname{logit} P(y_i = 1) = a + b \operatorname{logit} \hat{p}_i, \tag{4.21}$$

where a and b are estimated by logistic regression; a well-calibrated model has $b = 1$ and $a = 0$. A constant prediction equal to the event rate π has $B = \pi(1 - \pi)$, which is the reference value quoted with the results.

Clinical usefulness was assessed by decision curve analysis [17] and by sensitivity, specificity and predictive values at fixed risk thresholds. At a threshold t, where a participant is flagged if $\hat{p} \geq t$, the net benefit is

$$\mathrm{NB}(t) = \frac{\mathrm{TP}(t)}{n} - \frac{\mathrm{FP}(t)}{n} \cdot \frac{t}{1 - t}, \tag{4.22}$$

which counts true positives and penalises false positives by the odds of the threshold. It is compared with flagging everyone, $\mathrm{NB}_{\mathrm{all}}(t) = \pi - (1-\pi)\, t/(1-t)$, and with flagging no one ($\mathrm{NB} = 0$), for t from 0.02 to 0.60. Sensitivity is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, specificity $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, positive predictive value $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ and negative predictive value $\mathrm{TN}/(\mathrm{TN}+\mathrm{FN})$.

For the progression outcome, the mean absolute error $\mathrm{MAE} = \frac{1}{n}\sum_i |\Delta_i - \hat{\Delta}_i|$, the coefficient of determination $R^2 = 1 - \sum_i (\Delta_i - \hat{\Delta}_i)^2 / \sum_i (\Delta_i - \bar{\Delta})^2$ and the Spearman rank correlation were reported, each against the reference $\hat{\Delta} = 0$.

Uncertainty was quantified by $R = 1{,}000$ bootstrap resamples of the evaluation set. For a statistic θ, the 95% interval is $[\theta^*_{(0.025)}, \theta^*_{(0.975)}]$, the 2.5th and 97.5th percentiles of the resampled values $\theta^*_1, \ldots, \theta^*_R$. Differences between models were computed on the same resamples (paired bootstrap), so that the interval reflects the difference itself rather than two separate uncertainties. Performance was examined within subgroups defined by sex, age, race, education, APOE status and baseline diagnosis.

4.12.1 Survival analysis

As a check that does not exclude participants with incomplete follow-up, a Cox proportional hazards model [19] was fitted to the time to dementia diagnosis, with censoring at the last visit:

$$h(t \mid \mathbf{x}) = h_0(t) \exp\left(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}\right), \tag{4.23}$$

where h_0 is an unspecified baseline hazard and $\tilde{\mathbf{x}}$ the standardised features, so that e^{β_j} is the hazard ratio per standard deviation of feature j. The coefficients maximise the partial likelihood with a small L_2 penalty (0.05). Discrimination was measured by Harrell's concordance index, the proportion of comparable pairs in which the participant with the earlier event has the higher predicted hazard, and the cardiovascular contribution by the paired difference in concordance over 200 bootstrap resamples.

Chapter 5

PET Image Pipeline

This chapter describes a separate experiment on the 599 raw PET acquisitions. Its purpose is to establish whether the images can be processed into reliable quantitative measurements and used by an image model. It does not feed the dementia model, for the reason given in Section 5.6.

5.1 From DICOM series to static volumes

Each acquisition arrives as a dynamic DICOM series of between 188 and 4,134 files: the same volume repeated over several time frames, from about 20 scanner models. For every scan the stored values were converted to activity concentration, slices were grouped into frames and ordered spatially, and the frames whose mid-point fell inside the tracer's standard static acquisition window (for example 50–70 minutes after injection for florbetapir and PiB, and 90–110 minutes for florbetaben, with five minutes' tolerance) were averaged, each weighted by its duration:

$$V(\mathbf{r}) = \frac{\sum_{k \in \mathcal{W}} d_k \, V_k(\mathbf{r})}{\sum_{k \in \mathcal{W}} d_k}, \tag{5.1}$$

where $V_k(\mathbf{r})$ is the activity at voxel $\mathbf{r}$ in frame k, d_k the frame duration and $\mathcal{W}$ the set of frames in the window. The result is one static three-dimensional volume per scan. Between two and eight frames, covering 10 to 40 minutes of data, were averaged per scan (Figure 5.1). The raw files were only read; all outputs were written to a separate directory.

5.2 Spatial registration

Scans differ in matrix size, voxel spacing and head position, so the same voxel does not correspond to the same anatomical location across scans. No structural MRI was available, so a PET-only registration in two passes was used. In the first pass each scan was aligned to the MNI152 T1 template with a rigid transform (rotation and translation), using

mutual information as the similarity measure because PET and T1 images have different contrast. For a fixed image A and a moving image B resampled through the transform $\mathcal{T}$, the mutual information is

$$\mathrm{MI}(A, B \circ \mathcal{T}) = \sum_{a,b} p_{AB}(a,b) \log \frac{p_{AB}(a,b)}{p_A(a)\, p_B(b)}, \tag{5.2}$$

where p_{AB} is the joint histogram of intensities (32 bins, Mattes estimator) and p_A, p_B its marginals. It is largest when the intensities of one image best predict those of the other, without requiring them to be equal. The transform was optimised by gradient descent over three resolution levels (shrink factors 4, 2, 1), and each scan was smoothed with a Gaussian kernel ($\sigma = 3$ mm) for registration only; the unsmoothed image was resampled for analysis. The aligned scans of each tracer class were averaged to form a study-specific PET template, and in the second pass each scan was aligned to that template with a full affine transform, $\mathcal{T}(\mathbf{r}) = \mathbf{A}\mathbf{r} + \mathbf{t}$, with 12 parameters (nine in $\mathbf{A}$ for rotation, scaling and shear, three in $\mathbf{t}$ for translation). The determinant of $\mathbf{A}$ measures the volume scaling applied; it lay between 0.65 and 1.25 for all scans (Figure 5.1). Figure 5.2 shows the average of the registered scans for each tracer class, which is sharp only if the scans are well aligned. Aligning PET to a PET template is more robust than aligning it to an MRI template.

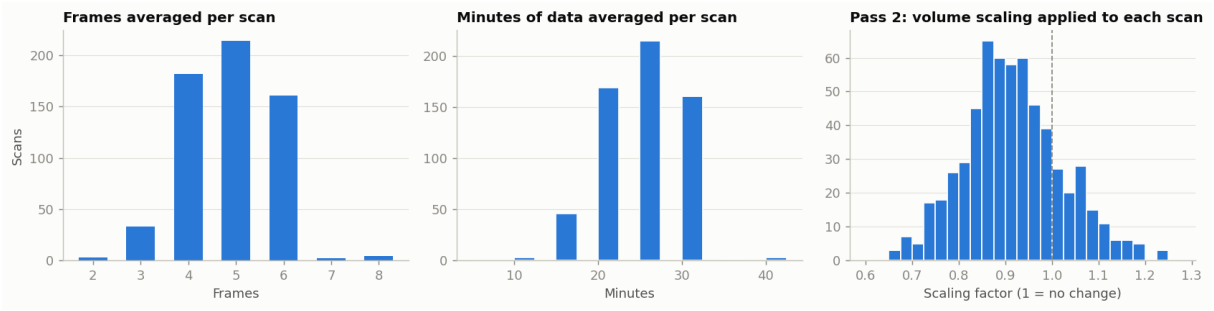


Figure 5.1: Quality control of the conversion and registration: number of frames and minutes of data averaged per scan, and the volume scaling applied by the affine registration (1 = no change).

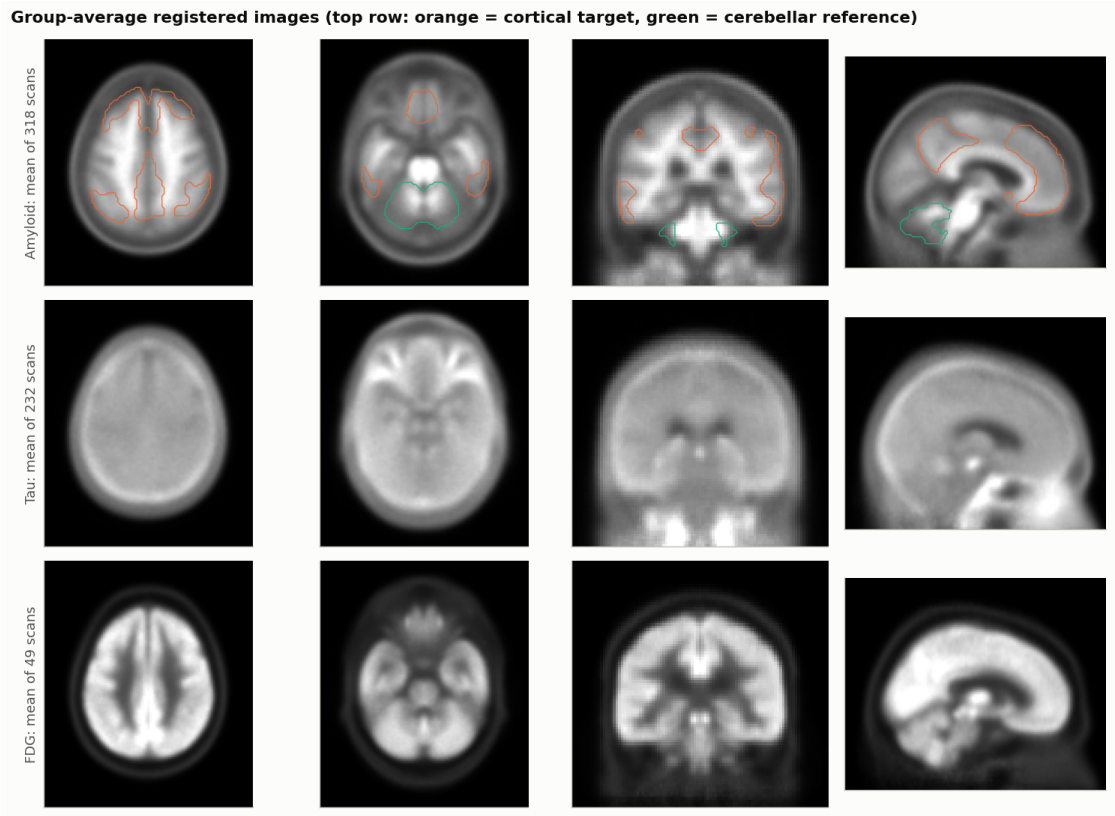


Figure 5.2: Average of all registered scans for each tracer class (amyloid, 318 scans; tau, 232; FDG, 49), in axial, coronal and sagittal sections. In the top row the orange outline is the Centiloid cortical target region and the green outline the cerebellar reference region. Only group averages are shown; no individual scan is reproduced.

5.3 Amyloid SUVR from the images

For each registered amyloid scan, the standardised uptake value ratio was computed as the mean uptake in the cortical target region divided by the mean uptake in the whole cerebellum, using the standard masks of the Centiloid project [12]:

$$\mathrm{SUVR} = \frac{\frac{1}{|\Omega_{\mathrm{ctx}}|}\sum_{\mathbf{r}\in\Omega_{\mathrm{ctx}}} V(\mathbf{r})}{\frac{1}{|\Omega_{\mathrm{cb}}|}\sum_{\mathbf{r}\in\Omega_{\mathrm{cb}}} V(\mathbf{r})}, \tag{5.3}$$

where Ω_{ctx} and Ω_{cb} are the voxels of the cortical and cerebellar masks. Because the activity scale cancels in the ratio, SUVR does not depend on the injected dose or body weight. NACC's PET core had computed an SUVR for the same scans with a different, MRI-based pipeline in native space. Agreement between the two therefore provides an independent test of the conversion and registration steps. Agreement was summarised by the Pearson and Spearman correlations, the least-squares line $\mathrm{SUVR}_{\mathrm{ours}} = a + b\,\mathrm{SUVR}_{\mathrm{NACC}}$, and the Bland–Altman limits of agreement $\bar{d} \pm 1.96\, s_d$, where $\bar{d}$ and s_d are the mean and standard deviation of the differences $d = \mathrm{SUVR}_{\mathrm{ours}} - \mathrm{SUVR}_{\mathrm{NACC}}$.

Table 5.1: Agreement between image-derived and NACC amyloid SUVR (318 scans).

Measure	Value
Pearson correlation	0.977
Spearman correlation	0.885
Regression line (ours on NACC)	$-0.16 + 1.10\,x$
Mean difference (ours minus NACC)	-0.043
95% limits of agreement	-0.18 to $+0.09$

Agreement was high for every tracer (Pearson r between 0.96 and 0.99 for the four tracers with known identity) and every scanner manufacturer. The lower Spearman correlation reflects the many amyloid-negative scans with similar low values, where small differences change the rank order.

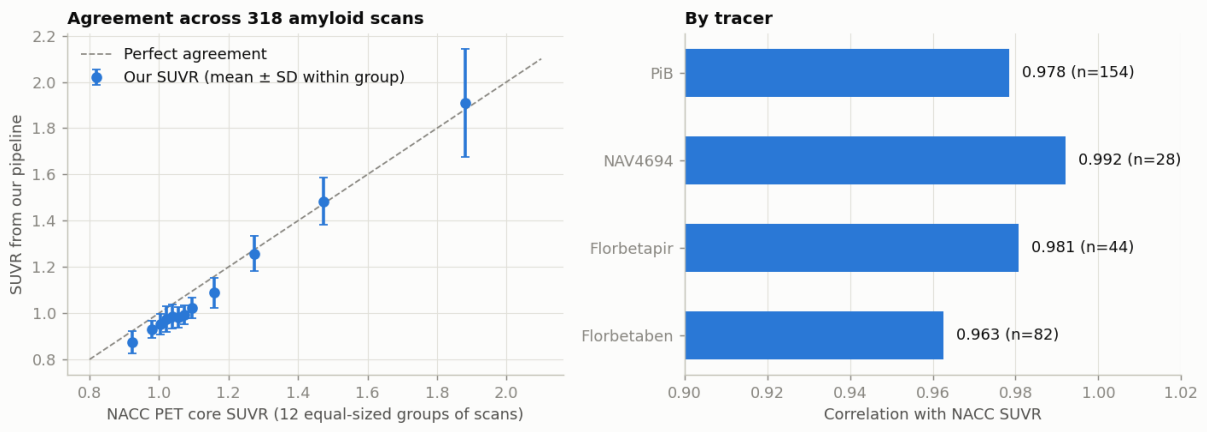


Figure 5.3: Image-derived amyloid SUVR against the value computed by the NACC PET core for the same scans.

5.4 Amyloid status classification

Three classifiers of amyloid status were compared on the 292 amyloid scans with a status label from the PET core (84 positive), using five-fold stratified cross-validation so that every scan was predicted by a model that had not seen it: (i) the cortical SUVR alone; (ii) a logistic regression on mean uptake in an $8 \times 8 \times 8$-voxel grid of blocks; and (iii) a small 3D convolutional neural network. The network has four stages of two $3 \times 3 \times 3$ convolutions with instance normalisation and ReLU, each stage halving the resolution, followed by global average pooling, dropout of 0.3 and a linear output, with 494,917 weights in total. Volumes were cropped to the head and resampled to $80 \times 96 \times 80$ voxels. Training used AdamW (learning rate 10^{-3}, weight decay 10^{-2}) with a one-cycle schedule, 40 epochs, batches of 8, and augmentation by left–right flips, shifts of up to three voxels and small intensity changes. Because positive scans are the minority, the cross-entropy

was weighted,

$$\mathcal{L} = -\frac{1}{n}\sum_{i}\Big[w_{+}\, y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\Big], \qquad w_{+} = \frac{1-\bar{y}}{\bar{y}}, \tag{5.4}$$

where $\bar{y}$ is the proportion of positive scans in the training folds. The regional logistic model used the mean uptake in each $8 \times 8 \times 8$ block as a feature, with a strong L_2 penalty ($C = 0.01$).

Table 5.2: Amyloid status classification (292 scans, 84 positive, cross-validated).

Method	AUROC
Cortical SUVR (one number)	0.982
Regional logistic model	0.971
3D CNN	0.960

The single SUVR value performed best (Figure 5.4).

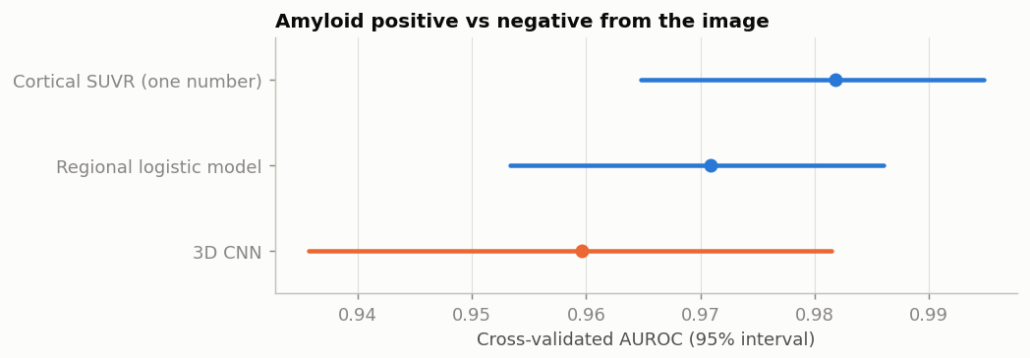


Figure 5.4: Cross-validated AUROC with 95% bootstrap intervals for the three amyloid-status classifiers.

This is expected: the reference label is itself derived by the PET core from a cortical uptake measure, so a scalar SUVR is close to the definition of the label, and a network trained on fewer than 300 scans has little opportunity to improve on it. The experiment shows that the image pipeline supports a working deep-learning model, not that a CNN is preferable for this task.

5.5 Grad-CAM

Gradient-weighted class activation maps [22] were computed for the network and averaged over 72 scans. If $\mathbf{A}^k$ is the k-th feature map of the last convolutional stage and s the output score, the map is

$$L_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_{k}\alpha_k \mathbf{A}^k\right), \qquad \alpha_k = \frac{1}{Z}\sum_{\mathbf{r}}\frac{\partial s}{\partial A^k_{\mathbf{r}}}, \tag{5.5}$$

where Z is the number of voxels in the feature map; the map is upsampled to the image size and divided by its maximum. The maps were diffuse: about 96% of the volume exceeded a quarter of the maximum activation, and only 1.7% of the most strongly activated voxels lay inside the cortical target region, against 3.8% expected by chance. The maps therefore do not show that the network attends to the cortical regions where amyloid accumulates, and they are not used as evidence of what the model has learned. Figure 5.5 shows the group-average images of amyloid-negative and amyloid-positive scans together with the mean Grad-CAM map: the positive scans show the expected higher cortical uptake, whereas the map is highest at the lower edge of the brain rather than in the cortical target region.

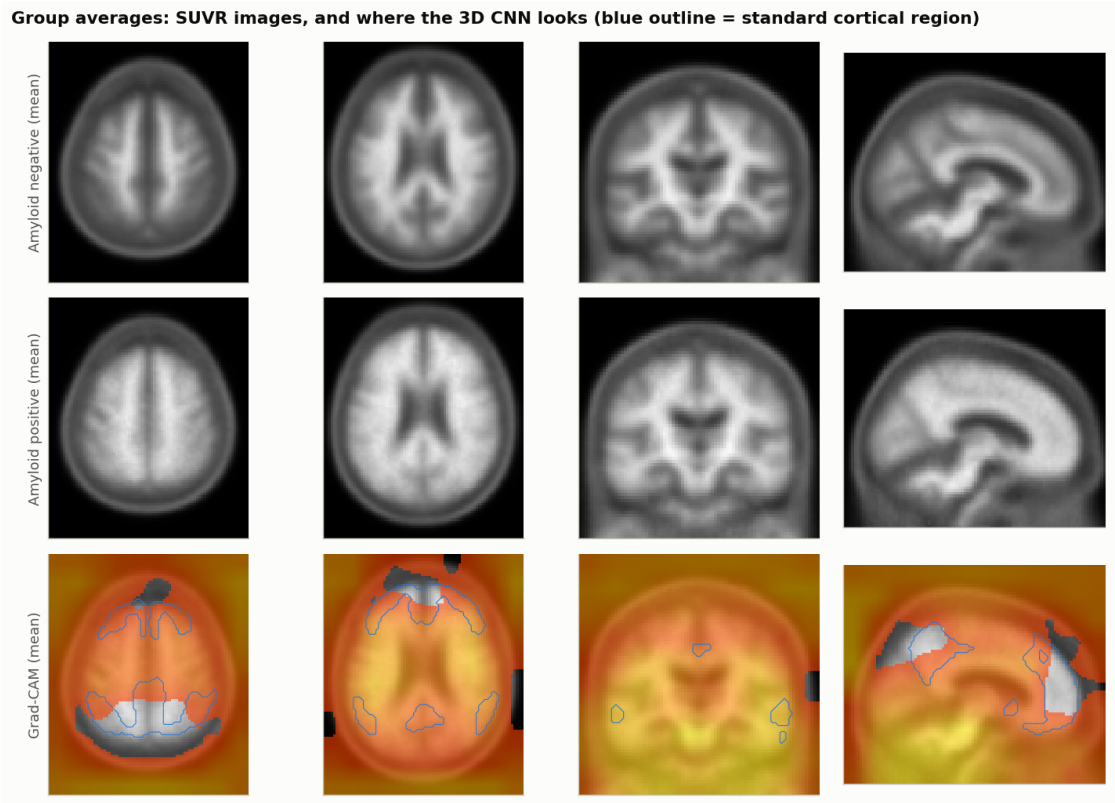


Figure 5.5: Group-average SUVR images of amyloid-negative and amyloid-positive scans (top two rows) and the mean Grad-CAM map of the 3D CNN (bottom row), in four sections. The blue outline marks the standard cortical target region.

5.6 Why images are not used to predict dementia

Of the 446 participants with an image, about 86% were cognitively normal at the time of the scan, follow-up after the scan is about two years, and only five later received a dementia diagnosis. No model can learn a reliable pattern from five events, so dementia prediction from images was not attempted. The quantitative PET information that is available for thousands of participants is used instead, through the late-fusion model described in Chapter 4.

Two further limitations apply. Registration without an MRI is less precise than the MRI-based pipelines used by PET cores. Tau images were converted and registered but not quantified, because a standard tau target mask comparable to the Centiloid masks was not available.

Chapter 6

Results

All results in this chapter are computed on participants who were not used for training: the internal test set (2,286 participants, 332 converters) and the nine held-out centres (4,064 participants, 545 converters). Intervals are 95% bootstrap intervals. Every number can be traced to a result file in the project repository.

6.1 Comparison of models

Table 6.1: Discrimination and calibration of the five models.

	Test set			Held-out centres	
Model	AUROC (95% CI)	AUPRC	Brier	AUROC	Brier
Logistic regression	0.936 (0.924–0.949)	0.721	0.066	0.945	0.060
Random forest	0.932 (0.919–0.944)	0.704	0.070	0.940	0.064
XGBoost	0.938 (0.925–0.948)	0.713	0.066	0.946	0.059
Multilayer perceptron	0.938 (0.926–0.949)	0.721	0.066	0.946	0.059
FT-Transformer	0.939 (0.927–0.949)	0.720	0.066	0.945	0.061

The five models perform almost identically, and their confidence intervals overlap almost completely (Table 6.1). The deep-learning models did not outperform logistic regression or gradient boosting on these data. For reference, always predicting the training event rate gives a Brier score of 0.124, so all models are substantially better calibrated than a constant prediction.

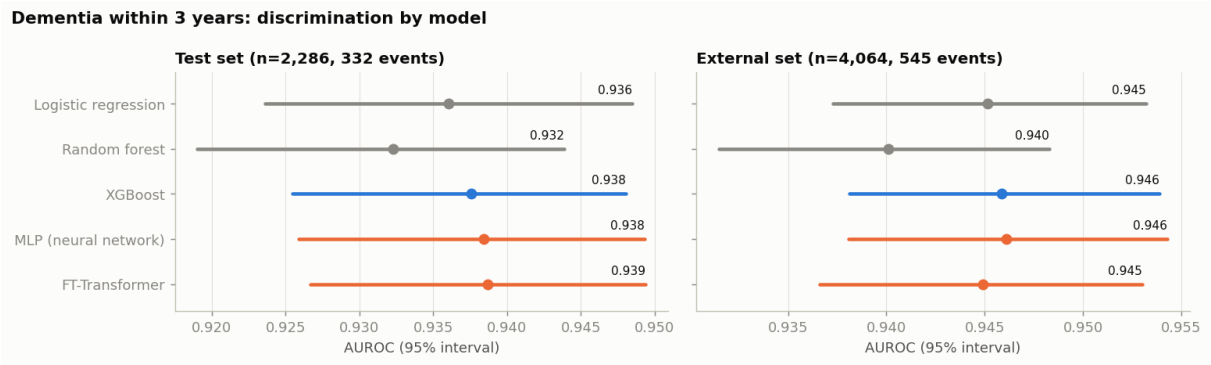


Figure 6.1: AUROC with 95% bootstrap intervals of the five models on the test set and the held-out centres (labelled "external set").

6.2 Performance by diagnosis at the index visit

The cohort combines cognitively normal participants, who rarely convert (1.6%), with participants with MCI, who often do (41.5%). Diagnosis alone therefore separates converters from non-converters well: a model given only the diagnosis reaches an AUROC of 0.84. Table 6.2 reports XGBoost separately within each group. The ablation, subgroup and missing-information analyses retrained XGBoost in separate runs, which give 0.937 for the whole test set instead of the 0.938 of Table 6.1; the difference of 0.0004 reflects the separate training runs and is far smaller than the confidence intervals.

Table 6.2: XGBoost by diagnosis at the index visit (participants / converters).

	Test set		Held-out centres	
Group	n / events	AUROC (95% CI)	n / events	AUROC (95% CI)
All not demented	2,286 / 332	0.937 (0.925–0.948)	4,064 / 545	0.946 (0.938–0.954)
Cognitively normal	1,411 / 24	0.884 (0.819–0.936)	2,624 / 37	0.876 (0.823–0.918)
MCI	732 / 289	0.835 (0.807–0.862)	1,291 / 496	0.846 (0.824–0.867)

Within MCI the AUROC is about 0.84, which is the more informative estimate of how well the model ranks risk among people who are already impaired. Within the cognitively normal group the AUROC is about 0.88, but it rests on only 24 and 37 events and the intervals are wide; with an event rate of 1.6% the average precision is only 0.15 to 0.19, so most participants flagged as high risk would not convert. Models trained only on normal participants (AUROC 0.876 and 0.881) or only on MCI participants (0.829 and 0.838) gave the same picture, so the group results are not an artefact of mixing the groups during training.

6.3 What drives the prediction

6.3.1 Feature categories

Table 6.3: Contribution of each feature category (XGBoost, test set unless stated).

Category	Alone	Gain added last (test)	Gain added last (held-out)
Cognitive tests	0.896	+0.022 (+0.015 to +0.029)	+0.019 (+0.013 to +0.024)
Clinical stage	0.897	+0.015 (+0.007 to +0.023)	+0.021 (+0.015 to +0.027)
Demographics	0.671	+0.010 (+0.006 to +0.015)	+0.009 (+0.006 to +0.013)
Genetics (APOE)	0.645	+0.003 (+0.002 to +0.004)	+0.001 (−0.000 to +0.002)
Cardiovascular/medical	0.626	+0.001 (−0.000 to +0.003)	+0.001 (+0.000 to +0.002)

Cognitive tests and clinical stage each reach an AUROC near 0.90 on their own and make the largest unique contributions (Table 6.3, Figure 6.2). Because these two categories overlap, removing either one costs little; removing both together, so that the model has no information about current cognition, lowers the AUROC from 0.937 to 0.746 on the test set (0.744 on the held-out centres). Within the normal and MCI groups the corresponding values are 0.783 and 0.638. The model therefore relies mainly on how impaired a person already is at the index visit. Because all these variables are recorded at the index visit and the outcome occurs later, this is not leakage, but it means the model is best described as estimating short-term progression given the current clinical stage.

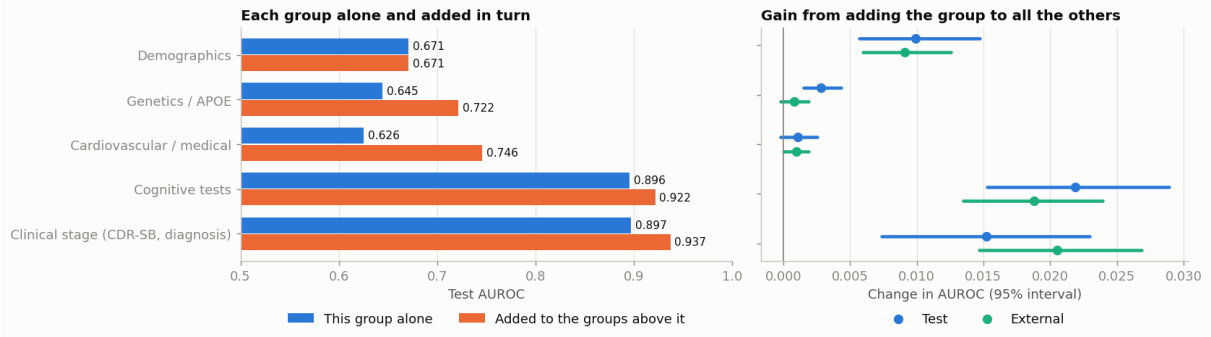


Figure 6.2: Left: AUROC of each category alone and added cumulatively. Right: gain in AUROC from adding each category to all the others.

6.3.2 Cardiovascular information (research question 2)

On its own, the cardiovascular category carries some signal (AUROC 0.626). Added to demographics and APOE only, before any cognitive information, it improved the AUROC by 0.024 on the test set (0.011 to 0.039), but by only 0.004 at the held-out centres (−0.006 to 0.015). Once cognitive information is present its contribution falls to about 0.001, with an interval that includes zero on the test set. Within the cognitively normal group the gain was 0.008 on the test set and 0.003 at the held-out centres, with intervals spanning

zero. In this cohort and prediction setting, therefore, cardiovascular and medical history added little incremental predictive value beyond current cognitive status. This finding does not imply that vascular health is unimportant for the brain: the participants enter at a median age of 71, mid-life exposure is not observed, and vascular damage that has already occurred is likely to be reflected in the cognitive scores.

6.3.3 Shapley explanations

Table 6.4: Share of mean absolute SHAP value by category (test set, 2,286 participants).

Category	Share
Clinical stage (CDR-SB, diagnosis)	40.8%
Cognitive tests	35.0%
Demographics	15.5%
Cardiovascular and medical	5.1%
Genetics	3.5%

The most important individual features were CDR-SB, MCI status, the memory score, age and the executive score. The ranking was stable across eight bootstrap refits (Spearman correlation with the main model 0.89 to 0.97; 8 to 10 of the top ten features shared). Some associations are not causal: lower BMI was associated with higher predicted risk, which is consistent with weight loss in the years before a dementia diagnosis rather than a protective effect of higher weight. The shares were similar within each diagnostic group (clinical stage 42% and 39%, cognitive tests 34% and 37% in normal and MCI participants; Figure 6.4).

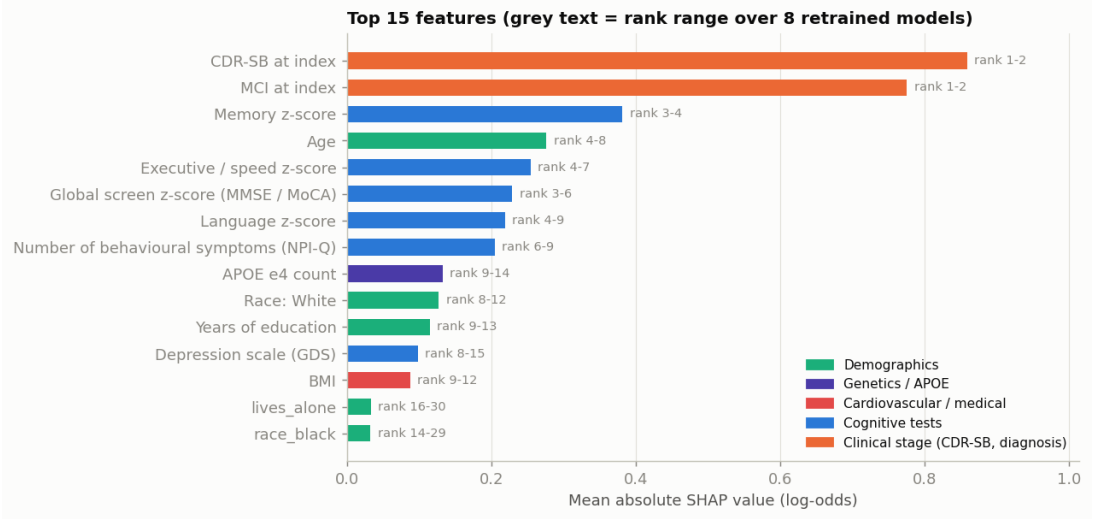


Figure 6.3: The fifteen features with the largest mean absolute SHAP value on the test set, coloured by category. The grey text gives the range of each feature's rank over eight models retrained on bootstrap samples.

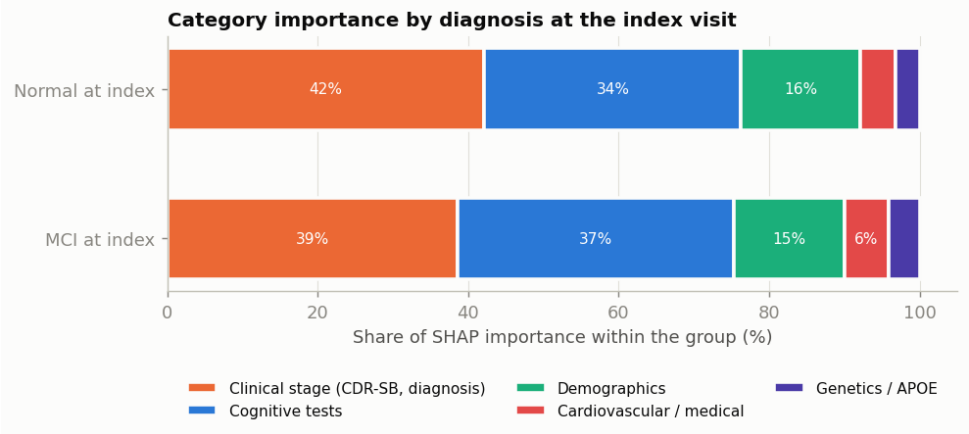


Figure 6.4: Share of SHAP importance by category within cognitively normal participants and participants with MCI at the index visit.

6.4 Cognitive Decline as the Outcome

Table 6.5 reports the secondary outcome, a rise of at least one point in CDR-SB within three years. The cohort is slightly larger than for dementia (20,355 participants, 5,035 events) because the outcome can be observed in people who decline without reaching dementia.

Table 6.5: XGBoost for CDR-SB worsening of at least one point within three years (participants / events).

	Test set		Held-out centres	
Group	n / events	AUROC (95% CI)	n / events	AUROC (95% CI)
All not demented	2,375 / 572	0.858 (0.841–0.874)	4,272 / 1,035	0.885 (0.875–0.896)
Cognitively normal	1,459 / 129	0.760 (0.714–0.805)	2,690 / 215	0.831 (0.806–0.856)
MCI	766 / 399	0.748 (0.712–0.779)	1,411 / 774	0.749 (0.723–0.776)

Figure 6.5 compares the two outcomes. Decline is harder to predict than dementia, as expected for an outcome that includes small changes in a rating scale. The cognitively normal group is much better represented here (129 and 215 events instead of 24 and 37), so the estimate of 0.76 to 0.83 in that group is more reliable than the corresponding dementia result. Logistic regression gave almost the same result (0.854 and 0.883). The pattern of contributions matched the dementia outcome: cognitive tests made the largest unique contribution (+0.029 on the test set), and the cardiovascular category added nothing measurable (+0.0004, interval −0.003 to +0.003; held-out centres 0.000).

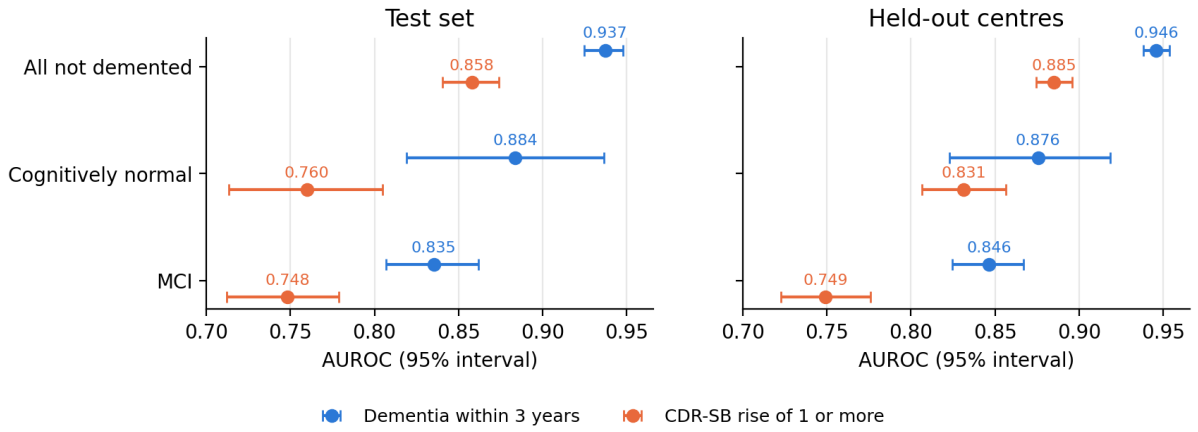


Figure 6.5: AUROC with 95% intervals of XGBoost for dementia within three years and for a rise of at least one point in CDR-SB within three years, by diagnosis at the index visit.

6.4.1 Predicted Change in CDR-SB

For the 1,863 test participants and 3,231 held-out participants with a visit near the three-year mark, the mean change in CDR-SB was 0.48 and 0.63 points, and 75% and 70% showed no change or an improvement. The regression model reduced the mean absolute error from 0.69 (predicting no change) to 0.60 on the test set and from 0.82 to 0.72 at the held-out centres, with R^2 of 0.23 and 0.18 and a Spearman correlation of 0.45 on both sets (Figure 6.6). Within MCI the error fell from 1.90 to 1.59 points on the test set and from 1.97 to 1.65 at the held-out centres. Within cognitively normal participants, whose scores rarely change, the model was no better than predicting no change (0.17 against 0.17 on the test set, 0.26 against 0.27 at the held-out centres). The progression estimate is therefore informative for people who are already impaired and adds little for people who are not.

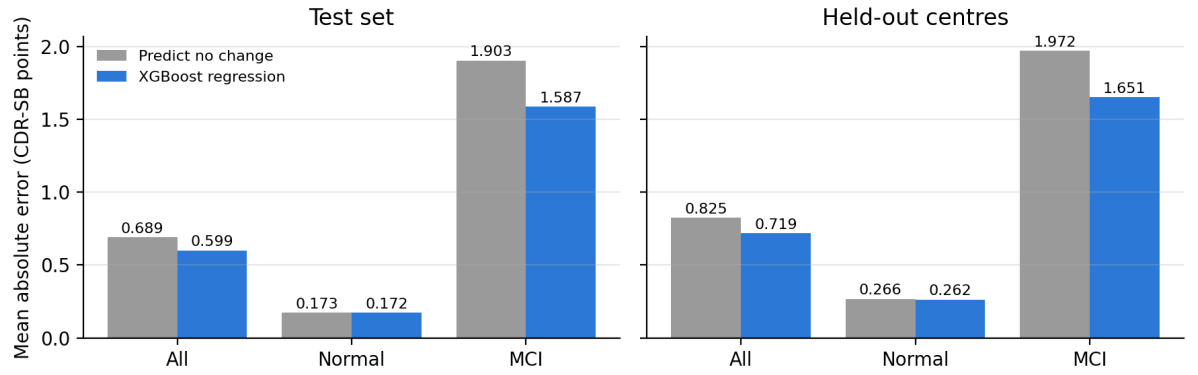


Figure 6.6: Mean absolute error of the predicted three-year change in CDR-SB, against the reference of predicting no change, by diagnosis at the index visit.

6.5 Modality-Encoder Fusion Network

The fusion network reached an AUROC of 0.934 (0.920–0.946) on the test set and 0.943 (0.935–0.951) at the held-out centres for dementia, against 0.937 and 0.946 for XGBoost; within MCI 0.831 and 0.837, and within cognitively normal participants 0.846 and 0.874. Its progression head reduced the mean absolute error from 0.72 to 0.65 on the test set. With modalities removed it behaved like the modality-dropout XGBoost model: without any cognitive information its AUROC was 0.72 and its Brier score 0.114, close to the specialist value.

The attention weights, intended as an explanation of how much the network relied on each modality, were nearly equal: between 18% and 22% for every modality, on both evaluation sets and in every diagnostic group, against 20% for perfectly uniform attention (Figure 6.7). They stayed almost equal even when the cognitive modalities were missing, because the learned missing tokens themselves carry the information. The attention weights therefore do not reveal which modality drives a prediction, in line with the wider observation that attention weights are not reliable explanations. The ablation results and SHAP values, which show that cognitive information dominates, remain the explanations used in this work. The fusion network demonstrates that the proposed architecture works and copes with missing modalities, but it offers no gain in accuracy over gradient-boosted trees on these data.

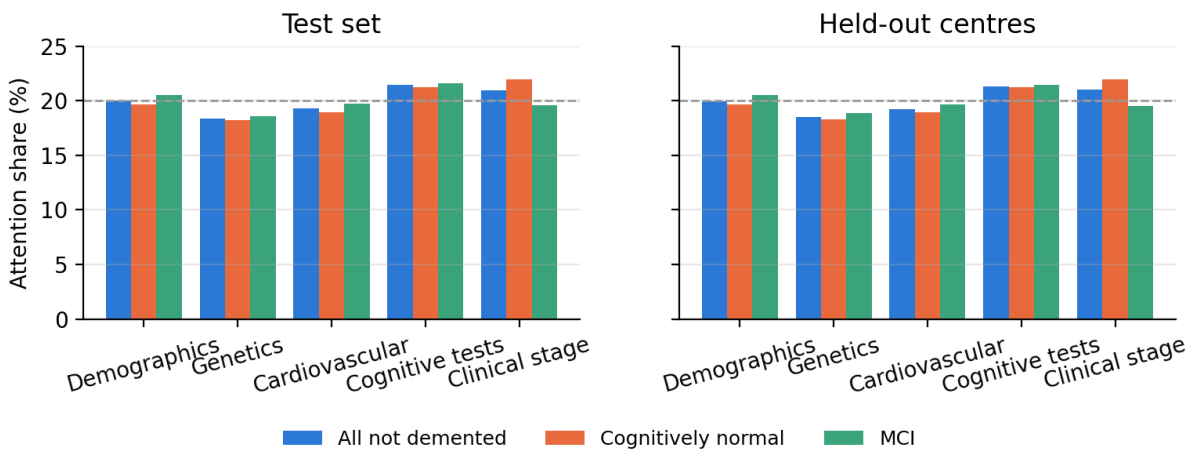


Figure 6.7: Mean attention share of each modality in the fusion network, by evaluation set and diagnosis at the index visit. The dashed line marks uniform attention (20%).

6.6 Longitudinal information

On landmark visits preceded by at least two earlier visits (test: 1,205 participants, 166 events; held-out centres: 2,254 participants, 326 events), adding the visit history produced only small changes (Table 6.6, Figure 6.8).

Table 6.6: Effect of earlier visits (paired gain over the landmark-only model).

Model	Test AUROC (gain)	Held-out AUROC (gain)
XGBoost, landmark visit only	0.919	0.953
XGBoost + trajectory features	0.922 (+0.003; −0.006 to +0.011)	0.956 (+0.003; −0.001 to +0.007)
GRU over the visit sequence	0.920 (+0.000; −0.011 to +0.010)	0.958 (+0.005; +0.001 to +0.008)

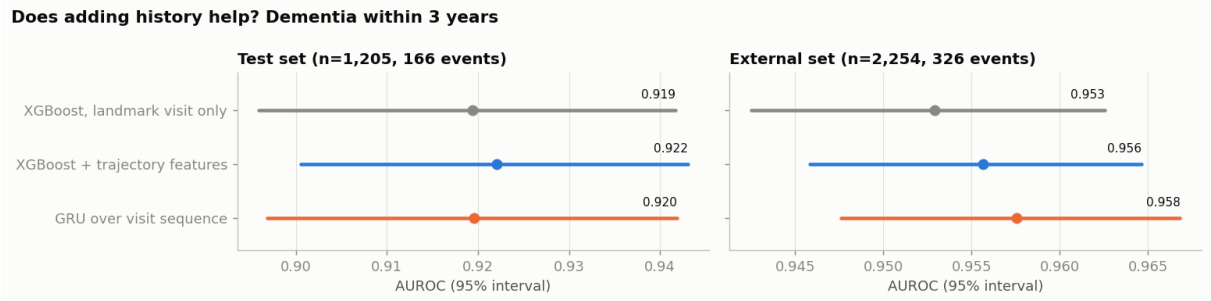


Figure 6.8: AUROC with 95% intervals of the three longitudinal models on landmark visits.

When the current scores are known, earlier scores add little to a three-year prediction. This does not show that longitudinal modelling is of no value; a longer horizon, an earlier disease stage or more frequent measurement could give a different result.

6.7 PET measures (research question 3)

On its own, amyloid status is strongly associated with progression: among participants with MCI in the held-out sets, about 34% of amyloid-positive participants received a dementia diagnosis within two years, compared with about 7% of amyloid-negative participants (36% and 9% under the strict scan-matching rule; Figure 6.9). Added to the clinical model, however, the PET measures did not produce a clear improvement under either matching rule. For example, with strict matching and a three-year horizon, the AUROC at the held-out centres was 0.939 without and 0.946 with amyloid (191 participants, 27 events; paired gain −0.007 to +0.024). The final late-fusion model scored 0.964 against 0.944 for the clinical model on 243 held-out scanned participants with 33 events, a gain of 0.020 whose interval (−0.009 to +0.066) includes zero. Figure 6.10 shows the gain for both matching rules and horizons with the test set and held-out centres pooled; every interval includes zero. A modest benefit cannot be excluded, but it could not be demonstrated with the follow-up currently available.

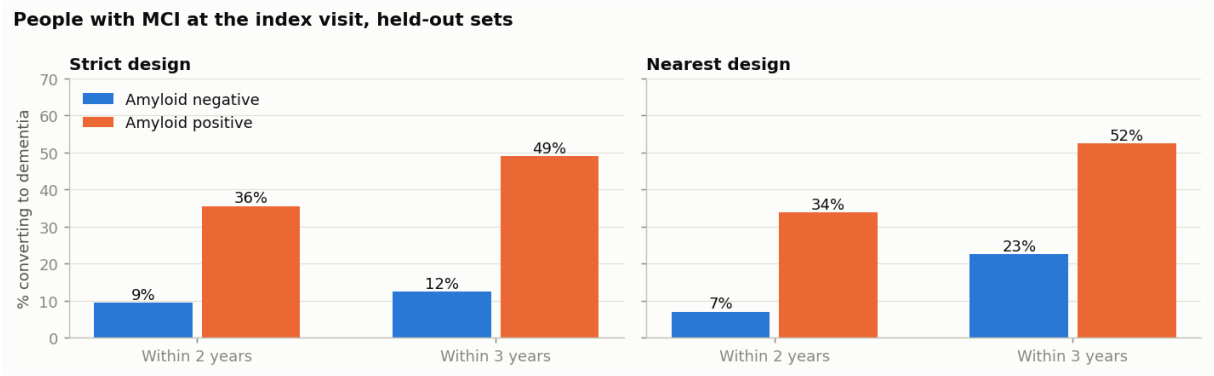


Figure 6.9: Percentage of participants with MCI at the index visit who converted to dementia within two and three years, by amyloid status, under the strict and nearest scan-matching rules (test set and held-out centres).

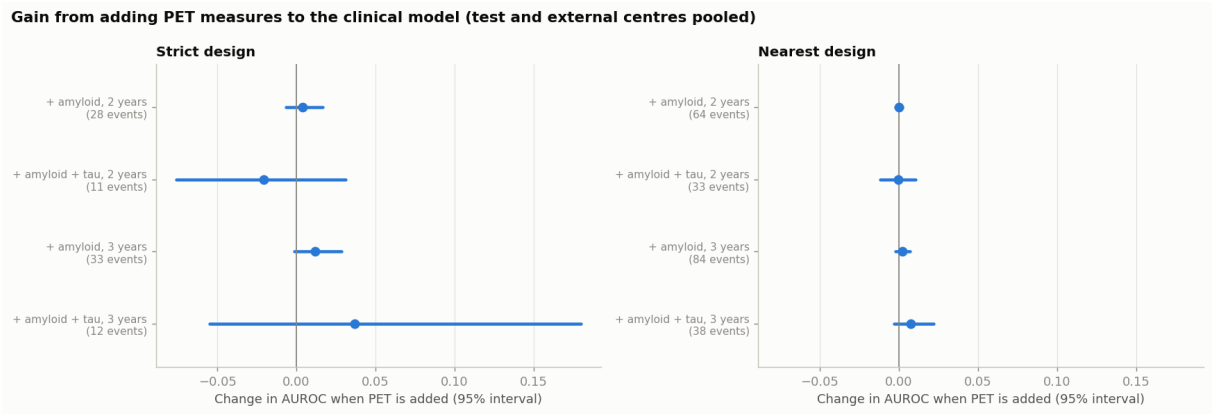


Figure 6.10: Change in AUROC when PET measures are added to the clinical model, with 95% intervals, test set and held-out centres pooled. The number of events is given for each comparison.

6.8 Validation

6.8.1 Subgroups

Table 6.7: XGBoost AUROC by subgroup.

Subgroup	Test (n / events)	Held-out centres (n / events)
Female	0.950 (1,338 / 162)	0.947 (2,507 / 272)
Male	0.919 (948 / 170)	0.941 (1,557 / 273)
Age under 65	0.976 (467 / 46)	0.958 (1,067 / 56)
Age 65–74	0.952 (963 / 99)	0.956 (1,463 / 191)
Age 75–84	0.904 (732 / 157)	0.937 (1,044 / 191)
Age 85 and over	0.852 (124 / 30)	0.888 (490 / 107)
White	0.933 (1,813 / 273)	0.943 (3,347 / 499)
Black	0.948 (304 / 31)	0.951 (471 / 26)
Education 12 years or less	0.926 (400 / 92)	0.920 (744 / 137)
Education over 16 years	0.951 (937 / 112)	0.960 (1,551 / 188)
APOE ε4 carrier	0.915 (762 / 150)	0.933 (1,402 / 277)

Discrimination was similar across sex, race and education (Figure 6.11). It was lower in the oldest participants and in APOE ε4 carriers, both groups with a higher baseline risk.

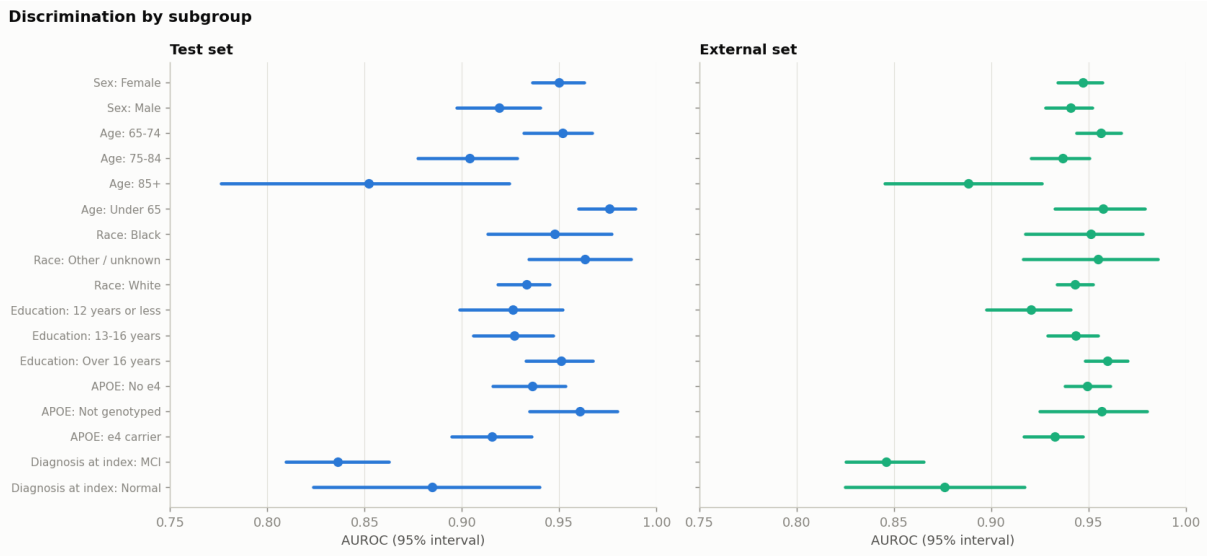


Figure 6.11: XGBoost AUROC with 95% intervals by subgroup on the test set and the held-out centres ("external set").

6.8.2 Calibration and clinical usefulness

On the test set the mean predicted risk (14.5%) equalled the observed rate (14.5%), with a calibration slope of 1.01 and intercept of 0.009. At the held-out centres the model over-predicted slightly (15.0% predicted, 13.4% observed; slope 1.10), as the calibration curves in Figure 6.12 show. Using a risk threshold of 20%, the model flagged 22% of

test participants with a sensitivity of 83%, specificity of 88%, positive predictive value of 55% and negative predictive value of 97% (held-out centres: 90%, 87%, 52% and 98%). Decision curve analysis showed a higher net benefit than treating everyone or no one across all thresholds examined, from 2% to 60%, on both evaluation sets (Figure 6.13).

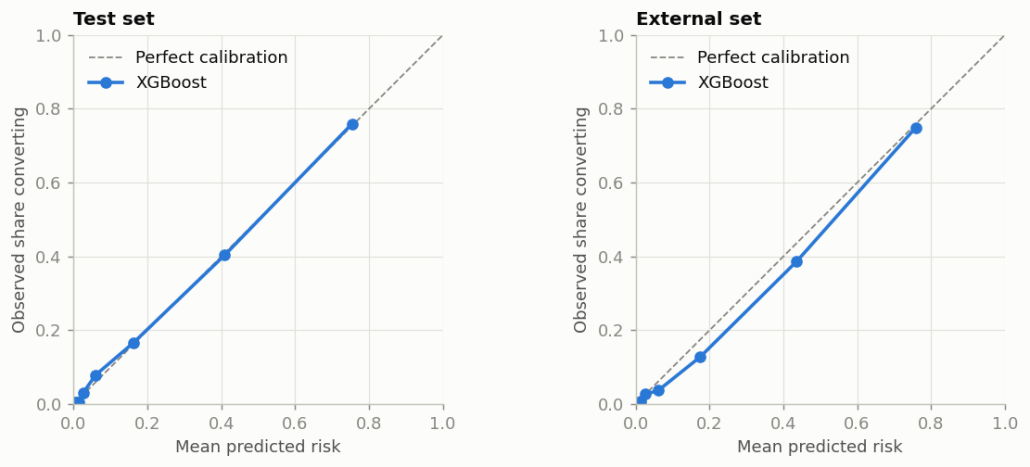


Figure 6.12: Calibration of XGBoost: observed share converting against mean predicted risk in groups of participants, on the test set and the held-out centres.

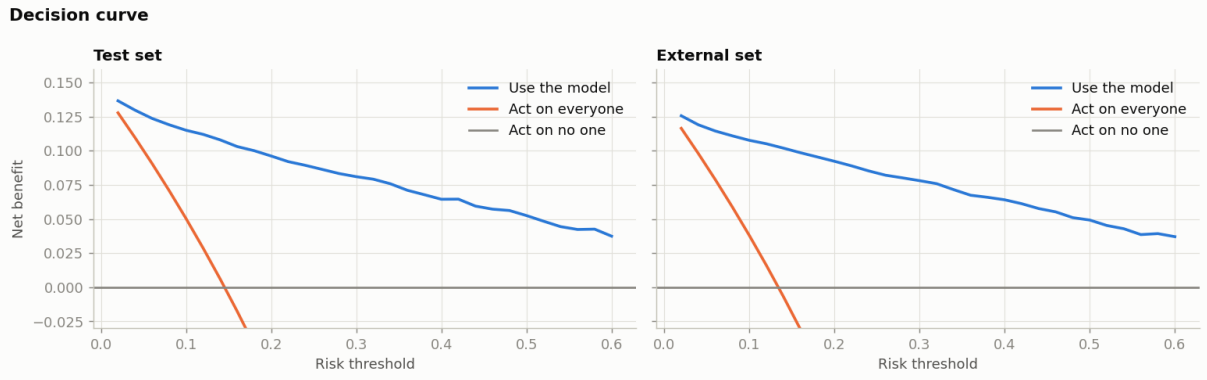


Figure 6.13: Decision curves: net benefit of acting on the model's prediction, of acting on everyone and of acting on no one, across risk thresholds.

6.9 Missing information

Table 6.8: Brier score when categories are missing (test set; lower is better).

Available information	Standard model	Modality dropout	Specialist model
Everything	0.066	0.067	0.066
No cognitive tests	0.082	0.077	0.080
No clinical stage	0.185	0.076	0.074
No cognitive information at all	0.441	0.115	0.111
Demographics only	0.444	0.120	0.118

A standard model given missing inputs at prediction time produced badly calibrated risks when cognitive information was absent, worse than predicting the base rate (Brier 0.124). The model trained with modality dropout stayed close to a specialist model retrained for each situation (Table 6.8), with almost no loss when all information is present (Figure 6.14). This model is used in the clinician interface.

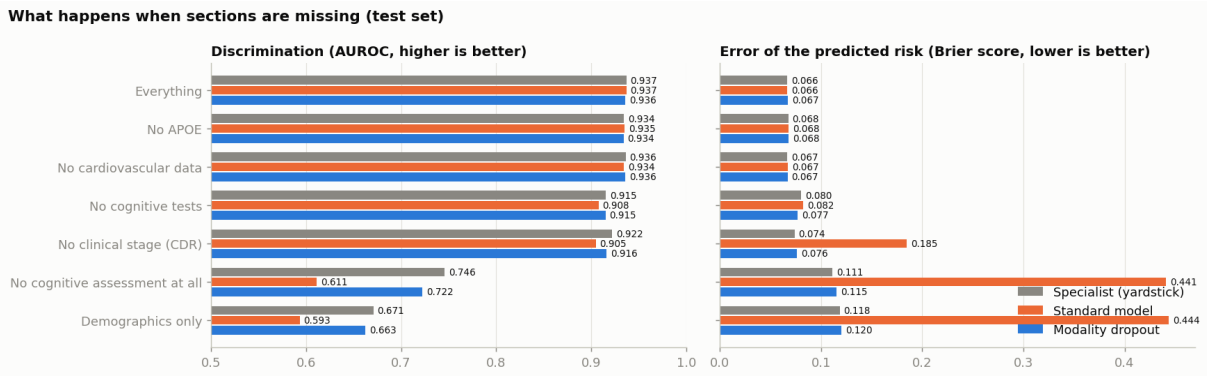


Figure 6.14: AUROC and Brier score on the test set when categories of information are missing, for the standard model, the modality-dropout model and a specialist model retrained for each pattern.

6.10 Sensitivity analyses

Censored participants. A Cox model fitted on all eligible participants, including those followed for less than three years, reached a concordance index of 0.891 on the test set and 0.903 at the held-out centres. Removing the cardiovascular category changed the concordance by 0.001 (−0.001 to 0.003) and 0.000 (−0.001 to 0.002). Hazard ratios per standard deviation were 1.37 for CDR-SB, 0.76 for the memory score, 1.40 for age and 1.20 for the APOE $\varepsilon 4$ count, while hypertension (0.995, 0.953–1.038), diabetes, stroke and systolic blood pressure had hazard ratios close to one (Figure 6.15).

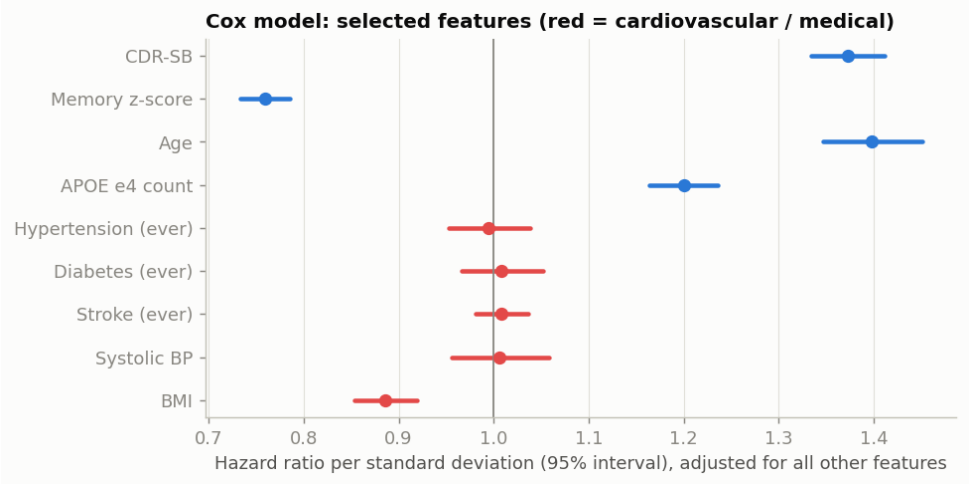


Figure 6.15: Hazard ratios per standard deviation with 95% intervals from the Cox model, adjusted for all other features. Cardiovascular and medical features are shown in red.

Reversed diagnoses. A clinical diagnosis can be revised. Of the 2,818 converters, 217 (7.7%) were recorded as not demented at a later visit. Excluding them, XGBoost reached an AUROC of 0.947 on the test set and 0.950 at the held-out centres (MCI: 0.839 and 0.851). The conclusions do not depend on reversed diagnoses.

Variables close to the diagnosis. Adding the daily-function questionnaire, independence, memory complaints and Alzheimer's medication increased the AUROC by 0.006 on the test set and 0.005 at the held-out centres. These variables were therefore left out of the main model with little loss of accuracy.

Chapter 7

System Design

The models are packaged as a research prototype for a clinician: an interface for entering a patient's information and seeing a risk estimate with its explanation, and an evidence assistant that answers questions from a fixed library of published papers. The prototype is not intended for clinical use and states this on every screen.

7.1 Architecture

Table 7.1: Components of the prototype.

Component	Role
Inference module (`scripts/inference.py`)	Converts raw entries into the 48 model features (for example a MoCA score into the global z-score), runs the modality-dropout model and the PET fusion model, computes SHAP values and checks the plausibility of the entries
API (`app/api.py`, FastAPI)	Serves the form definition, predictions and evidence-assistant answers
Interface (`app/web`, React)	The form and the results
Evidence assistant (`scripts/rag.py`)	Retrieval, guardrails and answer generation

7.2 Input form

The form is generated from the list of features the model was trained on, so it asks for exactly those inputs and nothing else. It has one section per category: demographics (required), cardiovascular history and measurements, the clinician's assessment (cognitive status and CDR-SB), test scores, genetics, and PET. The clinician marks each optional section as available or not. Empty fields are passed to the model as missing values; no default values are substituted. For each test the form shows the expected score of a healthy person of the same age, sex and education.

7.3 Output

The interface reports the estimated three-year risk next to the observed three-year conversion rate of training participants in the same age group and with the same cognitive status, which gives the clinician an empirical reference point. It lists which categories were used and which were missing, and shows the SHAP contributions of each category and of the most influential individual items. When no cognitive information is available, it states that the estimate is little more than an age-based average. Figure 7.1 shows the output for an example: a 72-year-old woman with 16 years of education, MCI, a CDR-SB of 2 and a MoCA score of 21, whose estimated risk of 39% sits next to the 40% observed among 1,283 training participants of the same age group and status.



Figure 7.1: The estimate panel of the prototype for the example patient: estimated risk, observed rate of similar participants, categories used, SHAP contributions by category, test scores relative to healthy people of the same age, sex and education, and the most influential single items.

7.4 Plain-Language Report

On request, the interface writes a short report for the clinician. It has five parts: a summary of the estimated risk next to the observed conversion rate of similar participants; the main reasons, built from the largest positive and negative SHAP contributions and phrased in plain terms (for example, "memory tests below expected for age, sex and education"); the reliability of the estimate, listing missing categories and any plausibility warnings; one or two quoted passages from the evidence library on the factors that drove the estimate, each with its source; and a fixed statement of the limits of the estimate. The report is assembled from templates filled with the person's own numbers, and the evidence is quoted rather than generated, so the report cannot state anything that the model or the literature did not provide. Figure 7.2 shows the report for the example patient.

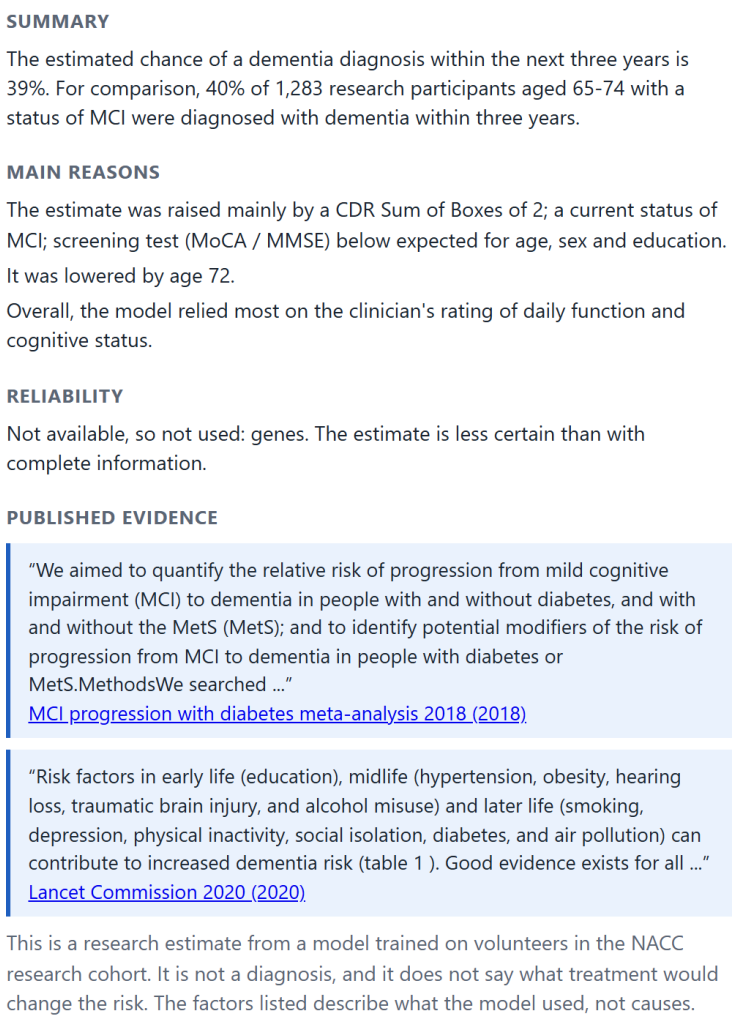


Figure 7.2: Plain-language report generated by the prototype for the example patient.

7.5 Plausibility checks

Contradictory entries can produce estimates that look precise but are meaningless. For example, a MoCA score of 0 entered together with a status of "normal" and a CDR-SB of 0 yields a low risk, because the model gives most weight to the clinician's rating. The interface therefore checks the entries against the ranges observed in training data before showing a result. Entries that contradict each other (test scores below the first percentile of cognitively normal participants combined with a normal status; a CDR-SB above the 99th percentile for the stated status; systolic pressure not above diastolic; implausible smoking duration) are flagged and the estimate is marked as unreliable. Unusual but possible entries (age outside the 1st to 99th percentile of the training cohort, very low education, discordant PET values) generate a note.

7.6 Behaviour checks

Twenty-eight automated checks test the deployed model through the same code path as the interface. They confirm that the form covers every trained feature and no other; that typical cognitively normal, impaired and MCI cases receive plausible, correctly ordered estimates (0.6%, 4.0% and 60.8%); that a worse CDR-SB, a worse cognitive status, lower test scores, more APOE $\varepsilon 4$ alleles or more behavioural symptoms never lower the estimate; that contradictory entries are flagged; and that missing categories are reported. Twenty-seven checks pass. The remaining check compares predicted and observed risk within age and diagnosis groups on held-out data: ten of eleven groups agree within seven percentage points, but for participants with MCI under the age of 65 the model predicts 35% against an observed 27%. This over-prediction is reported as a known limitation.

7.7 Evidence assistant

The evidence assistant is a retrieval-augmented generation system [23]. Its library consists of eleven open-access papers on dementia risk factors, vascular contributions, APOE, amyloid and tau PET and diagnostic criteria, divided into 447 passages. For a question, two rankings are computed, one by semantic similarity of sentence embeddings [24] and one by BM25 keyword relevance [25], and merged by reciprocal rank fusion [26]. For a question q and passage d, the semantic score is the cosine similarity of the normalised embeddings, $s_{\mathrm{sem}}(q,d) = \mathbf{e}_q^{\top}\mathbf{e}_d$, and the keyword score is

$$s_{\mathrm{BM25}}(q,d) = \sum_{w \in q} \mathrm{IDF}(w)\,\frac{f(w,d)\,(k_1+1)}{f(w,d) + k_1\bigl(1 - b + b\,|d|/\overline{|d|}\bigr)}, \qquad k_1 = 1.5,\ b = 0.75, \quad (7.1)$$

where $f(w,d)$ is the number of times word w occurs in d, $|d|$ the passage length and $\overline{|d|}$ the average length. If $\rho_{\mathrm{sem}}(d)$ and $\rho_{\mathrm{BM25}}(d)$ are the ranks of d under the two scores, the

passages are ordered by

$$s_{\mathrm{RRF}}(d) = \frac{1}{60 + \rho_{\mathrm{sem}}(d)} + \frac{1}{60 + \rho_{\mathrm{BM25}}(d)}, \tag{7.2}$$

which rewards passages ranked well by either method without having to put the two scores on a common scale. If the best semantic similarity is below a fixed threshold, $\max_d s_{\mathrm{sem}}(q, d) < 0.35$, the assistant reports that the library contains no evidence on the question. Otherwise a small instruction-tuned language model running locally writes an answer from the retrieved passages with numbered citations; if the answer does not cite the passages correctly, the assistant quotes them directly instead. Questions asking for advice about a particular person are answered with a notice that the system cannot give individual medical advice, followed by general published information only. No data leave the machine. Figure 7.3 shows an answer to a general question, given as direct quotations with their sources.

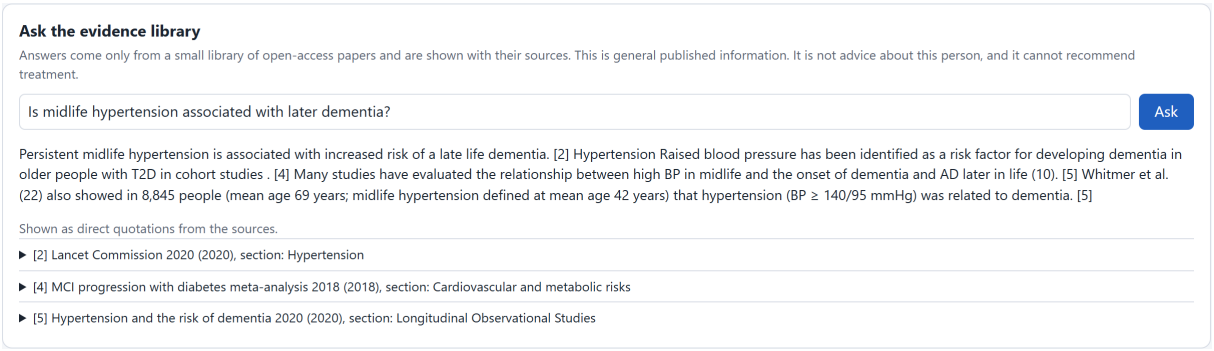


Figure 7.3: The evidence assistant answering "Is midlife hypertension associated with later dementia?" with quoted passages and numbered sources.

Table 7.2: Evaluation of the evidence assistant (52 questions).

Measure	Result
Correct source ranked first (hybrid retrieval)	32 of 32
Correct source ranked first (embeddings only / BM25 only)	28 of 32 / 31 of 32
Personal-advice questions flagged	8 of 8
General questions wrongly flagged as advice	0 of 6
Out-of-scope questions declined	6 of 6
Answers written with valid citations by the language model	22% (otherwise quoted)

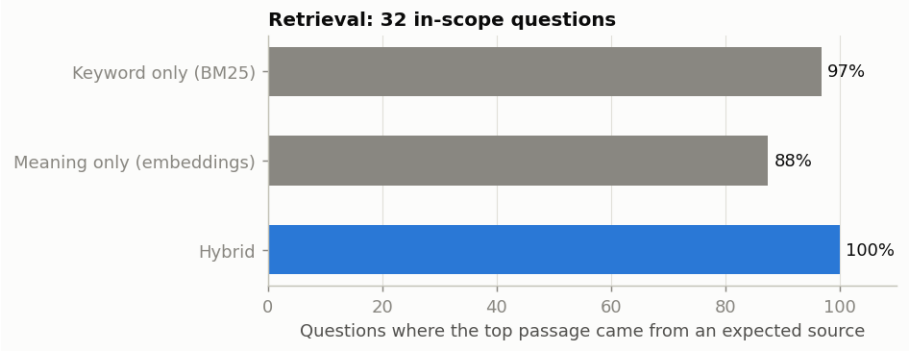


Figure 7.4: Share of the 32 in-scope questions for which the top-ranked passage came from an expected source, for keyword retrieval, embedding retrieval and the hybrid.

These results should be read with care. The library is small and the test questions were written by the project team, so retrieval is an easy test. The local language model is small and produced a properly cited answer for only about one question in five; direct quotation is the fallback and is the safer behaviour for medical content.

Chapter 8

Verification and Reproducibility

Clinical prediction studies are vulnerable to subtle errors that inflate performance: the same person in training and test data, information from the future in the features, mislabelled outcomes, or reported numbers that do not match the saved predictions. A separate verification program (`scripts/verify_results.py`) was written to test for these errors independently of the main pipeline. Each check recomputes a quantity in a second, simpler way or tests a property that must hold if the pipeline is correct.

8.1 Checks

Table 8.1: Independent verification checks and their results.

Group	Check	Result
A	Raw files unchanged since the first audit; all 599 image folders present	Pass
B	No participant in more than one partition; no held-out centre in the training data; saved split matches the modelling table	0 participants and 0 centres shared
C	Features at the first and third visit are identical when all later visits are deleted before the features are built	Identical for 1,500 participants × 67 features at each visit
C	No PET value is newer than the visit at which it is used	0 of 1,481 amyloid and 713 tau values
D	Outcome labels recomputed with an independent implementation	0 disagreements in 11,403 visits
D	No eligible index visit has a dementia diagnosis	0 of 134,943
E	With shuffled outcome labels, a retrained model falls to chance	AUROC 0.508
F	Reported AUROC values recomputed from the saved predictions	Largest difference 0 over 5 models and 2 evaluation sets
F	Cohort sizes in the reports match the modelling table	Match
G	Retraining from scratch with a new seed reproduces the result	Test 0.937, held-out 0.946 (reported 0.938, 0.946)
G	Gain from the cardiovascular category with a fresh seed	+0.0016
H	Cohort flow and PET index counts in the reports match the data	Match

All seventeen checks passed on the final pipeline (re-run on 23 October 2026).

8.2 Errors found and corrected

The checks found two genuine errors in an earlier version of the pipeline, both of which used information that would not have been available at the time of prediction:

1. Missing education and race at an early visit had been filled in from a later visit for about 0.1% of participants. Fixed characteristics are now carried forward only.

2. PET scans had been matched to the nearest visit, which for many participants was a visit a few weeks before the scan. A scan is now used only at a visit on or after its acquisition date.

After both corrections every model was retrained and every result regenerated. The main results changed only in the third decimal place. Earlier outputs were retained for comparison.

8.3 Reproducibility

All random seeds are fixed (2026 for the split, 0 to 2 for model training). Package versions are recorded in `requirements.txt`. The pipeline is run by a fixed sequence of scripts documented in the repository; given the NACC data, it regenerates every table in this thesis. Participant-level data and trained models are not distributed, in accordance with the data use agreement; all shared outputs are aggregates, and each report is checked automatically for participant identifiers before it is written.

Chapter 9

Discussion and Limitations

9.1 Interpretation of the Main Result

The headline AUROC of 0.94 must be read together with the composition of the cohort. Participants with MCI converted 25 times as often as cognitively normal participants, and the diagnosis alone reached an AUROC of 0.84. Much of the overall discrimination therefore comes from separating these groups, which a clinician already does. The more informative figures are the within-group results: about 0.84 among participants with MCI, and about 0.88 among cognitively normal participants, the latter based on few events. Removing all information about current cognition lowered the AUROC from 0.94 to 0.75. The model is therefore best described as estimating the short-term risk of progression given the current clinical stage, not as detecting future dementia in people without any signs of impairment.

Including CDR-SB and diagnosis as inputs is not leakage in the technical sense: both are measured at the index visit, and the outcome is a diagnosis made at a later visit. It does mean that the clinical label used as outcome and some of the inputs come from the same assessment process. Results without these inputs are reported for this reason.

9.2 The Cardiovascular Question

The central hypothesis of the original project was that cardiovascular information would improve prediction. In this cohort it added about 0.001 AUROC once current cognition was known, the survival analysis gave the same result, and the hazard ratios for hypertension, diabetes and stroke were close to one after adjustment. This does not contradict the epidemiological literature. Most of the evidence concerns midlife exposure or exposure accumulated over decades [2, 9], whereas NACC participants enter at a median age of 71 and their history is recorded largely as yes/no answers with single blood-pressure readings. Damage that vascular disease has already caused is likely to be reflected in the cognitive scores, which the model sees directly. The finding should therefore be stated

as limited incremental predictive value in this cohort and setting, not as evidence that vascular health is unimportant for the brain.

9.3 Comparison with Related Studies

The finding that cardiovascular information adds little once other information is known agrees with Aschwanden et al., who found cardiovascular factors less important than expected when evaluated jointly with other predictors [31]. It does not contradict the association studies [28, 29], which ask whether cardiovascular risk is related to later cognition rather than whether it improves a model that already measures current cognition. Our AUROC is higher than that of risk models without cognitive inputs, such as the UK Biobank model of Stern and Linial (0.773) [30], because our model includes current cognition and clinical stage; without them our model scored 0.75, in the same range. Like Yi et al. [32], we found CDR-SB to be the most important feature, although their task was diagnosis rather than prediction of future conversion.

9.4 Model Complexity

Logistic regression, random forest, XGBoost, a multilayer perceptron and an FT-Transformer performed within 0.007 AUROC of one another, with overlapping intervals. This agrees with comparisons on other tabular data [15]. With a few dozen informative features, most of them strongly related to the outcome, there is little structure left for a more flexible model to exploit. The modality-encoder fusion network proposed at the start of the project gave the same result, and its attention weights were nearly uniform, so they could not serve as explanations. The deep-learning idea that did prove useful was modality dropout, which addresses a practical problem (missing assessments) rather than raising discrimination.

9.5 PET

Amyloid status was strongly associated with progression in participants with MCI, in line with the biological definition of the disease [8], but no clear gain was shown when PET was added to the clinical model. The likely reason is limited follow-up: most scans are recent, and only a few dozen scanned participants with a known three-year outcome converted. The PET image experiment showed that the raw images can be processed into measurements that agree closely with the PET core's, but with five converters among the participants with images, image-based prediction of dementia was not attempted.

9.6 Limitations

1. **Cohort.** NACC is a research cohort recruited through memory centres and volunteers; participants are mostly White and highly educated. Absolute risks and performance may differ in other populations.

2. **Validation.** The held-out centres belong to the same study, use the same forms and were collected in the same period. This is site-level validation, not external validation on an independent cohort, and no temporal validation was performed.

3. **Censoring.** About half of the eligible participants were excluded because their three-year outcome was unknown. Participants with more severe impairment return less often, so the analysed cohort is somewhat healthier than all enrolled participants. The survival analysis that includes censored participants reached the same conclusions.

4. **Outcome.** The outcome is a clinical diagnosis, which can be revised; 7.7% of converters were later recorded as not demented. Excluding them did not change the results.

5. **Cognitively normal participants.** Only 24 test and 37 held-out converters were cognitively normal at the index visit, so performance in this group is uncertain.

6. **Calibration.** Risk was slightly over-predicted at the held-out centres and by about eight percentage points for participants with MCI younger than 65.

7. **Demographic inputs.** Race, ethnicity and living situation improved calibration across racial groups, most likely because of differences in recruitment, but they change individual estimates noticeably. Whether such inputs belong in a clinical tool is a question of policy as well as accuracy.

8. **Test reuse.** The same test set was used for several experiments. The held-out centres provide a second evaluation set, and the results agree.

9. **PET images.** Registration was performed without MRI, tau images were not quantified, and Grad-CAM maps were not informative.

10. **Evidence assistant.** The library contains eleven papers and the test questions were written by the team.

9.7 Ethical Considerations

The data were used under the NACC data use agreement. No participant-level information leaves the analysis machine, and every shared output is checked automatically for identifiers. The prototype is labelled as a research tool, does not diagnose, and does not

recommend treatment: the data record what happened to participants and cannot show what would have happened under a different treatment. The evidence assistant answers only from published sources and declines to give advice about an individual.

Chapter 10

Conclusion and Scope for Further Research

10.1 Conclusion

This thesis set out to predict cognitive decline over the next three years for people without dementia at a clinic visit, to establish which kinds of information contribute to that prediction, and to explain each estimate in a form a clinician can check. Using the NACC database under a person-level split with nine held-out centres, the answers to the seven research questions are as follows.

RQ1. Progression to dementia within three years can be predicted with an AUROC of 0.94 in the combined cohort and 0.95 at held-out centres; within MCI the AUROC is about 0.84, and within cognitively normal participants about 0.88 with wide uncertainty. Cognitive decline measured as a one-point rise in CDR-SB is predicted with an AUROC of 0.86 (0.89 at held-out centres), 0.75 within MCI and 0.76 to 0.83 within cognitively normal participants, and the predicted change in CDR-SB is informative mainly for people who are already impaired.

RQ2. Cardiovascular and medical history adds about 0.001 AUROC for dementia and nothing measurable for CDR-SB decline once current cognition is known.

RQ3. Amyloid and tau PET measures did not produce a demonstrable gain over the clinical model with the follow-up currently available.

RQ4. Neither deep tabular models nor the modality-encoder fusion network performed better than logistic regression or gradient-boosted trees.

RQ5. Earlier visits added between 0.000 and 0.005 AUROC beyond the current visit.

RQ6. A model trained with modality dropout remained close to specialist models when whole categories of information were missing, whereas a standard model became badly calibrated.

RQ7. Each estimate can be explained with Shapley values, checked for implausible inputs, and summarised in a plain-language report with quoted published evidence.

Most of the predictive information lies in the current cognitive and clinical state. The results are robust to the exclusion of participants with unknown outcomes, to reversed diagnoses and to retraining with different seeds, and the pipeline passed seventeen independent verification checks after two sources of information leakage had been found and corrected. Alongside the models, the work produced a PET image pipeline whose amyloid measurements agree with the PET core's (correlation 0.98), and a research prototype that checks the plausibility of its inputs and supports its output with a plain-language report and an evidence assistant restricted to published sources.

10.2 Scope for Further Research

10.2.1 External and Temporal Validation

The model should be evaluated on an independent cohort with different recruitment and on participants enrolled after the training period.

10.2.2 Cognitively Normal Participants and Longer Horizons

A model built specifically for cognitively normal participants, with a five- or ten-year horizon, would have more events and more room for cardiovascular and PET information to contribute.

10.2.3 Richer Cardiovascular Exposure

Repeated blood-pressure measurements over many visits, treatment history and imaging markers of small vessel disease could capture vascular burden better than yes/no history.

10.2.4 PET with Longer Follow-up

As more scanned participants reach three or more years of follow-up, the added value of amyloid and tau PET can be tested with adequate power, and MRI-based processing of the images can improve quantification.

10.2.5 Evidence Assistant

A larger, curated library, an independently written evaluation set and a stronger local language model would make the assistant more useful.

10.3 Final Remarks

The main finding of this work is a negative one, but it is supported by a controlled design and independent checks. In this cohort, the information that best predicts dementia within three years is the person's current cognitive and functional state. Long-term factors such as vascular history and genetics appear to be reflected in that state and add little independent information to a short-term prediction.

Appendix A

Feature List

Table A.1: The 48 model features and their source columns in the NACC clinical table.

Category	Feature	Source and construction
Demographics	age	NACCAGE
	female	NACCSEX = 2
	education (years)	EDUC, carried forward
	race (White, Black, Asian)	NACCNIHR, three indicators
	Hispanic	NACCHISP
	lives alone	NACCLIVS = 1
Cardiovascular	hypertension, hypercholesterolaemia, diabetes, myocardial infarction, atrial fibrillation, heart failure, angioplasty, bypass, pacemaker, other cardiovascular disease (10)	Form A5 (HYPERTEN, HYPERCHO, DIABETES, CVHATT, CVAFIB, CVCHF, CVANGIO, CVBYPASS, CVPACE, CVOTHR); form D2 where A5 is blank; "ever" carried forward
	stroke, TIA	NACCSTYR, NACCTIYR, with CBSTROKE, CBTIA
	ever smoked, years smoked	TOBAC100, SMOKYRS
	systolic and diastolic BP, heart rate, BMI	BPSYS, BPDIAS, HRATE, NACCBMI, latest measurement
	pulse pressure	systolic minus diastolic
	vascular risk count	hypertension + hypercholesterolaemia + diabetes + smoking
	medication classes and count (5)	NACCAHTN, NACCLIPL, NACCDBMD, NACCAC, NACCAMD
Cognitive tests	global score	z of MOCATOTS, NACCMMSE
	memory score	z of CRAFTVRS, CRAFTDVR, LOGIMEM, MEMUNITS, UDSBENTD
	attention score	z of DIGFORCT, DIGBACCT, DIGIF, DIGIB
	executive score	z of TRAILA, TRAILB, WAIS
	language score	z of ANIMALS, VEG, UDSVERFC, BOSTON, MINTTOTS
	visuospatial score	z of UDSBENTC
	tests not completed	count of codes 96 / 996
	depression	NACCGDS
	behavioural symptoms	count of twelve NPI-Q items
Clinical stage	CDR-SB	CDRSUM
	MCI, impaired not MCI	NACCUDSD = 3, = 2
Genetics	APOE $\varepsilon 4$ count, not genotyped	NACCNE4S
	family history	NACCFAM

Appendix B

Implementation and Reproducibility Notes

B.1 Software

Python 3.11 with the package versions listed in `requirements.txt`: NumPy 1.26, pandas 2.3, scikit-learn 1.6, XGBoost 3.0, PyTorch 2.8, SHAP 0.48, lifelines 0.30, SimpleITK 2.5, pydicom 3.0, sentence-transformers 6.1, FastAPI 0.116 and React 18 for the interface.

B.2 Pipeline

With the NACC files placed under `data/raw/`, every result is regenerated by running the scripts below in order.

1. Cohort and features: `audit_data.py`, `build_dataset.py`.
2. Dementia models: `train_tabular.py`, `train_by_baseline_group.py`, `train_longitudinal.py`, `explain.py`.
3. PET and validation: `train_pet.py`, `validate.py`, `train_final.py`.
4. Decline outcomes and fusion network: `train_decline.py`, `train_fusion_network.py`.
5. Sensitivity analysis: `sensitivity_confirmed.py`.
6. PET image pipeline: `pet_convert.py`, `pet_register.py`, `pet_suvr.py`, `train_pet_image.py`.
7. Evidence assistant: `rag_build.py`, `rag_eval.py` (retrieval and answering in `rag.py`).
8. Checks: `verify_results.py`, `check_clinical_behaviour.py`.
9. Figures: the twelve notebooks in `notebooks/` redraw the project figures from the saved results, and `thesis/make_figures.py` and `thesis/make_result_figures.py` draw the remaining thesis figures.

B.3 Seeds

Data split: 2026. Model training: 0, 1 and 2, averaged. Image cross-validation: 0.

B.4 Data Access

Participant-level data and trained models cannot be distributed under the NACC data use agreement. All reported results are aggregates and are stored in the `reports` directory of the project repository.

References

[1] G. Livingston, J. Huntley, A. Sommerlad, D. Ames *et al.*, "Dementia prevention, intervention, and care: 2020 report of the Lancet Commission," *The Lancet*, vol. 396, no. 10248, pp. 413–446, 2020.

[2] C. Sierra, "Hypertension and the risk of dementia," *Frontiers in Cardiovascular Medicine*, vol. 7, p. 5, 2020.

[3] A. H. Hainsworth, H. S. Markus, and J. A. Schneider, "Cerebral small vessel disease, hypertension, and vascular contributions to cognitive impairment and dementia," *Hypertension*, vol. 81, no. 1, pp. 75–86, 2024.

[4] F. Cao, F. Yang, J. Li, W. Guo *et al.*, "The relationship between diabetes and the dementia risk: a meta-analysis," *Diabetology & Metabolic Syndrome*, vol. 16, no. 1, p. 101, 2024.

[5] C. M. Wolfe, N. F. Fitz, K. N. Nam, I. Lefterov, and R. Koldamova, "The role of APOE and TREM2 in Alzheimer's disease—current understanding and perspectives," *International Journal of Molecular Sciences*, vol. 20, no. 1, p. 81, 2018.

[6] A. Chandra, P.-E. Valkimadi, G. Pagano, O. Cousins *et al.*, "Applications of amyloid, tau, and neuroinflammation PET imaging to Alzheimer's disease and mild cognitive impairment," *Human Brain Mapping*, vol. 40, no. 18, pp. 5424–5442, 2019.

[7] A. Leuzy, K. Chiotis, L. Lemoine, P.-G. Gillberg *et al.*, "Tau PET imaging in neurodegenerative tauopathies—still a challenge," *Molecular Psychiatry*, vol. 24, no. 8, pp. 1112–1134, 2019.

[8] C. R. Jack, J. S. Andrews, T. G. Beach, T. Buracchio *et al.*, "Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's Association Workgroup," *Alzheimer's & Dementia*, vol. 20, no. 8, pp. 5143–5169, 2024.

[9] R. Wang, Y. Gao, N. Ee, F. Akinyede *et al.*, "Cumulative blood pressure and risk of dementia and cognitive decline: a systematic review and meta-analysis," *The Journal of Prevention of Alzheimer's Disease*, vol. 13, no. 4, p. 100500, 2026.

[10] K. Pal, N. Mukadam, I. Petersen, and C. Cooper, "Mild cognitive impairment and progression to dementia in people with diabetes, prediabetes and metabolic syn-

drome: a systematic review and meta-analysis," *Social Psychiatry and Psychiatric Epidemiology*, vol. 53, no. 11, pp. 1149–1160, 2018.

[11] L. Lorenzini, A. Maranzano, S. Ingala, L. E. Collij *et al.*, "Association of vascular risk factors and cerebrovascular pathology with Alzheimer disease pathologic changes in individuals without dementia," *Neurology*, vol. 103, no. 7, p. e209801, 2024.

[12] W. E. Klunk, R. A. Koeppe, J. C. Price *et al.*, "The Centiloid Project: standardizing quantitative amyloid plaque estimation by PET," *Alzheimer's & Dementia*, vol. 11, no. 1, pp. 1–15, 2015.

[13] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[14] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[15] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting deep learning models for tabular data," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[16] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734.

[17] A. J. Vickers and E. B. Elkin, "Decision curve analysis: a novel method for evaluating prediction models," *Medical Decision Making*, vol. 26, no. 6, pp. 565–574, 2006.

[18] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457–481, 1958.

[19] D. R. Cox, "Regression models and life-tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 34, no. 2, pp. 187–202, 1972.

[20] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[21] S. M. Lundberg, G. Erion, H. Chen *et al.*, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[22] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 618–626.

[23] P. Lewis, E. Perez, A. Piktus *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[24] N. Reimers and I. Gurevych, "Sentence-BERT: sentence embeddings using Siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019, pp. 3980–3990.

[25] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, 2009.

[26] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.

[27] S. A. Alaka, S. C. Ngan, M. Shookoni, R. E. MacPherson *et al.*, "Enhancing the validity of CAIDE dementia risk scores with resting heart rate and machine learning: an analysis from the National Alzheimer's Coordinating Center across all races/ethnicities," *Alzheimer's & Dementia*, vol. 21, no. 8, p. e70442, 2025.

[28] R. Jia, Q. Wang, H. Huang, Y. Yang *et al.*, "Cardiovascular disease risk models and dementia or cognitive decline: a systematic review," *Frontiers in Aging Neuroscience*, vol. 15, p. 1257367, 2023.

[29] R. Song, H. Xu, C. S. Dintica, K.-Y. Pan *et al.*, "Associations between cardiovascular risk, structural brain changes, and cognitive decline," *Journal of the American College of Cardiology*, vol. 75, no. 20, pp. 2525–2534, 2020.

[30] A. Stern and M. Linial, "Integrative machine learning approach to risk prediction for dementia and Alzheimer's disease," *GeroScience*, vol. 48, no. 2, pp. 3007–3023, 2026.

[31] D. Aschwanden, S. Aichele, P. Ghisletta, A. Terracciano *et al.*, "Predicting cognitive impairment and dementia: a machine learning approach," *Journal of Alzheimer's Disease*, vol. 75, no. 3, pp. 717–728, 2020.

[32] F. Yi, H. Yang, D. Chen, Y. Qin *et al.*, "XGBoost-SHAP-based interpretable diagnostic framework for Alzheimer's disease," *BMC Medical Informatics and Decision Making*, vol. 23, no. 1, p. 137, 2023.

[33] S. El-Sappagh, J. M. Alonso, S. M. R. Islam, A. M. Sultan *et al.*, "A multilayer multimodal detection and prediction model based on explainable artificial intelligence for Alzheimer's disease," *Scientific Reports*, vol. 11, no. 1, p. 2660, 2021.

[34] S. Weintraub, L. Besser, H. H. Dodge *et al.*, "Version 3 of the Alzheimer Disease Centers' neuropsychological test battery in the Uniform Data Set (UDS)," *Alzheimer Disease & Associated Disorders*, vol. 32, no. 1, pp. 10–17, 2018.